

DIGITAL LITERACY

MICROSOFT EXCEL

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	MODULE	PERIOD	COST
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5.			
6.	MS Excel	8 Days	UGX200,000
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		8 Days	UGX200,000

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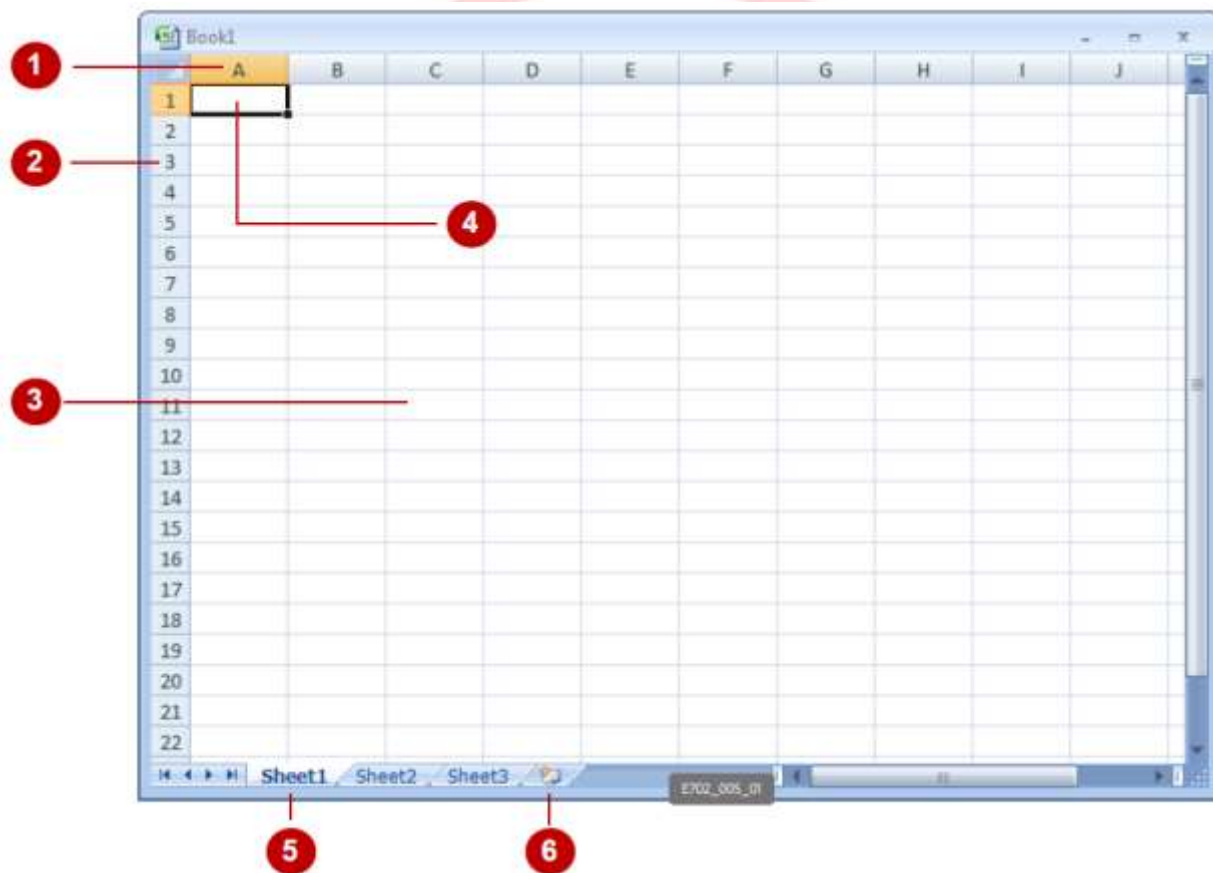
DAY #1

UNDERSTANDING WORKBOOKS

In Microsoft Excel the data you enter, whether it consists of numbers, text, or formulas, is stored in a file known as a workbook.

Workbooks are just like huge electronic books with pages (or sheets) that have been ruled into columns and rows.

Before using Excel, it is helpful to know what the various parts and elements that make up a workbook are.



1. A worksheet (or page) in a workbook contains 16,384 columns that are labelled using letters of the alphabet. The first column in a worksheet is labelled column A, while the last is labelled XFD
2. A worksheet (or page) in a workbook contains 1,048,576 rows that are labelled using numbers from 1 to 1,048,576
3. Where a column and row intersect we get what is known as a cell. You enter your data into these cells. Each cell in a worksheet can hold up to 32,767 characters – although it would be unrealistic to ever push it this far. Cells are referred to by their column and row

labels. For example, in the screen above the cell we are pointing to is C11 – this reference is known as the cell address and is most important as it is frequently used in commands and formulas

4. When you start typing something, you want it to appear somewhere in the worksheet. As a consequence, when the Status Bar shows Ready mode, at least one cell in the worksheet will be highlighted – this is known as the active cell. In the screen above, the active cell is cell A1 – notice that the column label and the row label also appears colored to indicate the active cell. You can have more than one active cell – when this occurs you have what is known as a range
5. A workbook (as you would expect) is made up of pages known as worksheets. You can have as many sheets in a workbook as your computer resources can accommodate. As a default, a new blank workbook normally has 3 worksheets labelled Sheet1, Sheet2, and Sheet3. Of course these labels are pretty boring and meaningless and can be changed to something more relevant
6. The Insert Worksheet button here will insert another worksheet into the current workbook should you need it

NAVIGATING IN A FILE

Arrow Keys	Move one cell to the right, left, up or down
Tab	Move one cell to the right
Ctrl + Home	To beginning file
Ctrl + End	To end of typed information
Home	Beginning of a line
End	End of a line
Page Down	Down one screen
Page Up	Up one screen
F5	To a specific page
Scroll bars	Appear at the right and on the bottom of the screen. You may click the scroll arrows, drag the scroll box or click the scroll bar to move through the document.

TYPING TEXT OR NUMBERS INTO A WORKSHEET

Generally, when you start a new spreadsheet project, the first task is to enter some headings into rows and columns.

To type anything into a worksheet you need to make the cell into which you wish to enter the data active.

This can be done in a number of ways but the most common is to click in it first before typing.

	A	B	C	D	E	F	H	I
1	STUDENT NAME	MATHS	ENGLISH	BIOLOGY	PHYSICS	CHEMISTRY	TOTAL MARKS	
2								
3	ANURAG KUMAR	87	57	77	63	87	371	
4	SAPTARSHI MONDAL	98	88	58	85	90	419	
5	SARTHAK GHOSH	85	95	45	90	81	396	
6	NISCHAY	32	62	39	98	62	293	
7	AKASH SHARMA	66	46	73	66	76	327	
8	DEEPESH	72	12	53	70	72	279	
9	PRATEEK	56	76	94	66	80	372	
10	PRATISH	98	66	43	87	44	338	
11	SHIVANI	92	52	62	91	77	374	
12	SHRUTI	59	49	72	49	34	263	
13	SHREYA	47	60	31	87	17	242	
14								
15								

TYPING SIMPLE FORMULAS IN A WORKSHEET

The whole idea behind Excel is to get it to perform calculations.

In order for it to do this you need to type formulas in the worksheet. Usually these formulas reference existing numbers, or even other formulas, already in the worksheet using the cell addresses of these numbers rather than the actual value in them.

Formulas must be typed beginning with an equal sign (=).

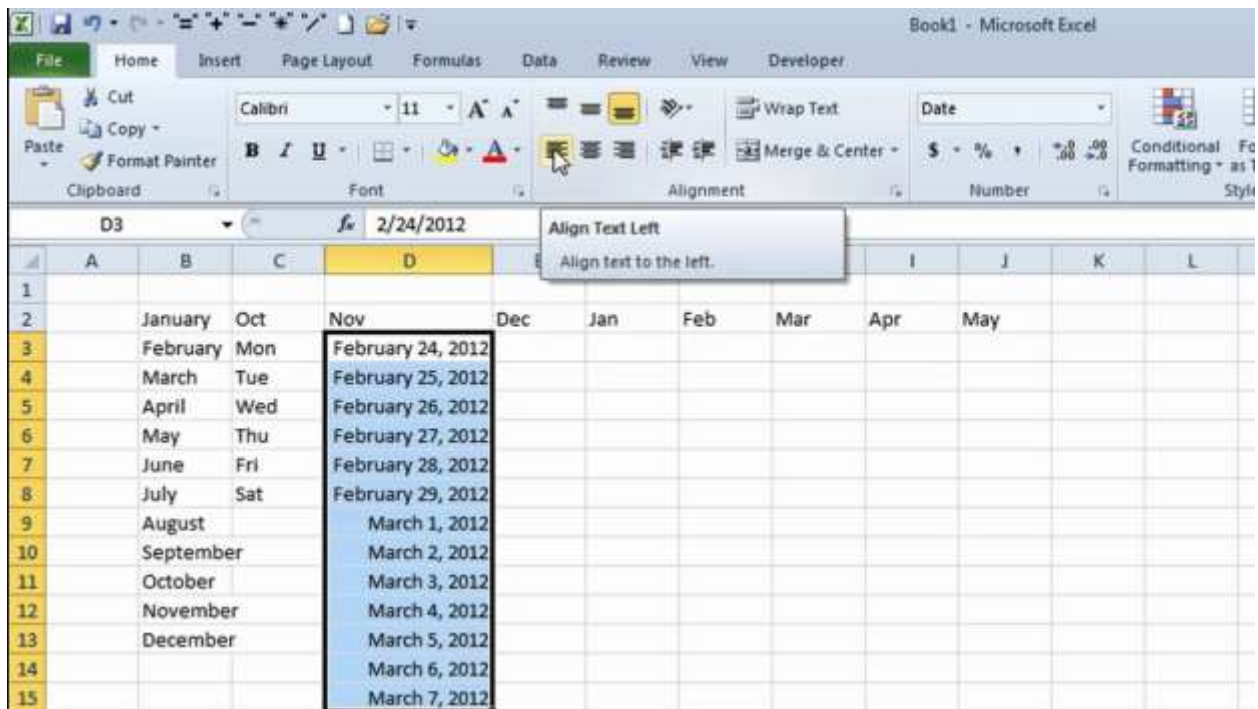
=SUM(A1:A5)	=AVERAGE(A1:A5)	=MAX(A1:A5)
=COUNT(A1:A5)	=MIN(A1:A5)	=IF(A1>55,"P","F")
=LOOKUP	=HLOOKUP	=VLOOKUP

FILLING A SERIES

A series refers to a sequence of ordered entries in adjacent cells, such as the days of the week or months of the year.

The fill technique can be used to create these in a worksheet for you, reducing the amount of time taken for data entry, and ensuring that the spelling is correct.

Excel provides days and months as special built-in series that you can access.



EXERCISE

1. A _____ consists of a grid made from columns and rows?

- A. Cell
- B. Labels
- C. Spreadsheet
- D. Formulas

Correct Answer

C. Spreadsheet

Explanation

A spreadsheet consists of a grid made from columns and rows. It is a digital tool used for organizing and analyzing data. The grid structure allows users to input data into individual cells, which can be organized and manipulated using formulas and functions. The columns and rows provide a systematic way of organizing and referencing data within the spreadsheet.

2. What is not a basic part of a spreadsheet?

- A. Formula

- B. Cell
 - C. Row
 - D. Column
-

Correct Answer

A. Formula

Explanation

A formula is not a basic part of a spreadsheet because it is a more advanced feature that allows users to perform calculations and manipulate data within the spreadsheet. While cells, rows, and columns are fundamental components of a spreadsheet, formulas enable users to perform complex calculations by referencing other cells, performing mathematical operations, and applying functions. Formulas are essential for automating calculations and data analysis in spreadsheets, but they are not considered a basic part of a spreadsheet as they require a deeper understanding and usage of spreadsheet software.

3. Which of the following data can be put inside a cell?

- A. Row
 - B. Column
 - C. Constant
 - D. None
-

Correct Answer

C. Constant

Explanation

A constant can be put inside a cell in a spreadsheet. A constant refers to a fixed value that does not change. In a spreadsheet, cells are used to store and manipulate data. Constants are often used to represent values that remain constant throughout the spreadsheet, such as mathematical constants or fixed values used in calculations. Placing a constant inside a cell allows for easy referencing and use of the constant value in formulas and calculations throughout the spreadsheet.

4. What is the symbol for an addition?

- A. -
 - B. +
 - C. =
-

D. ?

Correct Answer

B. +

Explanation

The symbol for addition is "+". It is commonly used to indicate the operation of combining two or more numbers to find their total sum. The plus sign is widely recognized and used in mathematical equations and calculations.

5. All formulas must begin with what?

A. =

B. +

C. -

D. /

Correct Answer

A. =

Explanation

All formulas must begin with the equal sign (=). The equal sign is used to indicate that the following characters are part of a formula and not just regular text. It tells the spreadsheet software that a calculation or function is about to be performed. Without the equal sign, the characters would be treated as text and not as a formula.

6. A fast way to add up this column of numbers is to click in the cell below the numbers and then:

A. Click Subtotals on the Data menu.

B. View the sum in the formula bar.

C. Click the AutoSum button on the Standard toolbar, then press ENTER.

Correct Answer

C. Click the AutoSum button on the Standard toolbar, then press ENTER.

Explanation

Clicking the AutoSum button on the Standard toolbar and then pressing ENTER is a fast way to add up the column of numbers. This feature automatically selects the range of cells above the active cell and inserts the SUM function in the formula bar, which calculates the sum of

the selected cells. Pressing ENTER confirms the calculation and displays the sum in the active cell.

7. Say that you want to paste a formula result — but not the underlying formula — to another cell. You would copy the cell with the formula, then place the insertion point in the cell you want to copy to. What next?

- A. Click the Paste button on the Standard toolbar.
 - B. Click the arrow on the Paste button on the Standard toolbar, then click Formulas.
 - C. Click the arrow on the Paste button on the Standard toolbar, then click Values.
-

Correct Answer

C. Click the arrow on the Paste button on the Standard toolbar, then click Values.

Explanation

After copying the cell with the formula, you would place the insertion point in the cell you want to copy to and then click the arrow on the Paste button on the Standard toolbar. From the options that appear, you would select "Values" to paste only the result of the formula without the underlying formula itself.

8. How do you change column width to fit the contents?

- A. Single-click the boundary to the left of the column heading.
 - B. Double-click the boundary to the right of the column heading.
 - C. Press ALT and single-click anywhere in the column.
-

Correct Answer

B. Double-click the boundary to the right of the column heading.

Explanation

To change the column width to fit the contents, you need to double-click the boundary to the right of the column heading. This action will automatically adjust the width of the column to accommodate the widest content within it.

9. There are three worksheets with every new workbook. You can change that automatic number if you want to.

- A. True
 - B. False
-

Correct Answer

A. True

Explanation

In Microsoft Excel, by default, every new workbook contains three worksheets. However, it is possible to change this automatic number of worksheets if desired. Therefore, the statement "There are three worksheets with every new workbook" is true.

10. ##### means:

- A. You've entered a number wrong.
 - B. You've misspelled something.
 - C. The cell is not wide enough.
-

Correct Answer

C. The cell is not wide enough.

Explanation

This answer suggests that the term "#####" is used to indicate that the cell in question is not wide enough to display the entire content. This could be because the content entered in the cell is too long and exceeds the width of the cell, causing the displayed content to be truncated with the "#####" symbol.

11. To add a new row, click a cell in the row immediately above where you want the new row.

- A. True
 - B. False
-

Correct Answer

B. False

Explanation

To add a new row, you need to click a cell in the row immediately below where you want the new row, not above. Therefore, the correct answer is False.

12. Which key do you press to group two or more nonadjacent worksheets?

- A. CTRL.
 - B. SHIFT
 - C. ALT
-

Correct Answer

B. CTRL

13. Which formula can add the all the numeric values in a range of cells, ignoring those which are not numeric, and place the result in a different cell?

- A. Count
 - B. Average
 - C. Sum
-

Correct Answer

C. Sum

Explanation

The formula that can add all the numeric values in a range of cells, ignoring those which are not numeric, and place the result in a different cell is the SUM formula. This formula is specifically designed to add up numerical values and can exclude non-numeric values from the calculation. It is a commonly used function in spreadsheet programs to calculate the total sum of a range of cells.

14. Is it possible to insert an image from a file into an Excel spreadsheet?

- A. Yes
 - B. No
-

Correct Answer

A. Yes

Explanation

It is possible to insert an image from a file into an Excel spreadsheet. This can be done by going to the Insert tab, selecting the Picture option, and then choosing the image file from the file explorer. The image will then be inserted into the spreadsheet and can be resized or positioned as needed.

15. Can an Excel spreadsheet be used as the "data source" for a Word Mail Merge ?

- A. Yes
 - B. No
-

Correct Answer

A. Yes

Explanation

An Excel spreadsheet can be used as the "data source" for a Word Mail Merge. This means that the data from the Excel spreadsheet can be merged into a Word document, allowing for personalized and customized documents to be created. The Excel spreadsheet acts as the source of data, providing the information that is needed to populate the Word document. This functionality is commonly used for creating letters, labels, and other documents that require personalized information for each recipient.

16. On an Excel sheet the active cell is indicated by ____.

- A. A dark wide border
 - B. A dotted border
 - C. A blinking border
 - D. None of the above
-

Correct Answer

A. A dark wide border

Explanation

The active cell on an Excel sheet is indicated by a dark wide border. This border helps to differentiate the active cell from the other cells on the sheet, making it easier for the user to identify and work with the selected cell. The dark wide border stands out and provides a visual indicator of the current active cell, allowing the user to input or manipulate data in that specific cell.

17. A formula and a function are the same thing.

- A. True
 - B. False
-

Correct Answer

B. False

Explanation

While formulas and functions are related concepts in spreadsheet software like Excel or Google Sheets, they are not the same thing. A formula is a general term for an expression that performs a calculation or operation on data within a spreadsheet. A function, on the other hand, is a specific type of formula that is pre-defined by the software to perform a particular task, such as summing a range of cells (SUM function) or calculating an average (AVERAGE function). So, all functions are formulas, but not all formulas are functions.

18. In order to multiply items in Excel you would use:

- A. ^
 - B. @
 - C. *
 - D. #
-

Correct Answer

C. *

Explanation

To multiply items in Excel, the asterisk (*) symbol is used. This symbol represents the multiplication operation in Excel.

19. The formula = ((A2+B5) * 5% is valid

- A. True
 - B. False
-

Correct Answer

B. False

Explanation

The given formula is not valid because it is missing a closing parenthesis after B5. The correct formula should be ((A2+B5)*5%).

20. If the cells in a Microsoft Excel worksheet contain the following values: A1 = 90 A2 = 85 A3 = 80 A4 = 75 A5 = 75 What formulae should you use to calculate the average of these values?

- A. =SUM(A1:A5)
 - B. =AVERAGE(A1:A5)
 - C. =MEDIAN(A1:A5)
 - D. =MODE(A1:A5)
-

Correct Answer

B. =AVERAGE(A1:A5)

Explanation

The AVERAGE function in Microsoft Excel is used to calculate the mean (or average) of a group of numbers. When you apply the AVERAGE function to a range of cells, Excel performs two main steps:

- Sum of the Values: Excel first adds up all the numbers in the specified range.
- Division by Count of Values: Then, it divides the total sum by the number of values in the range.

21. If cells: A1=90 A2=85 A3=80 A4=75 A5=75 What will be your formula if you are going to get the total?

- A. =SUM(A1:A5)
- B. =AVERAGE(A1:A5)
- C. =MEDIAN(A1:A5)
- D. =MODE(A1:A5)

Correct Answer

- A. =SUM(A1:A5)

Explanation

The SUM function in Microsoft Excel is specifically designed to add up all the numbers in a given range of cells. It is one of the most basic and frequently used functions in Excel, allowing users to quickly calculate the total of multiple values.

When you apply the SUM function to a range of cells, Excel performs the following steps:

- Identifies the Range: It identifies all the cells within the specified range (in this case, A1 to A5).
- Adds Up the Values: It then adds the values contained in each of those cells.

22. How would you Recognize the Cell Address ?

- A. By Name Box
- B. By Formula Bar
- C. By Ribbon

Correct Answer

- A. By Name Box

Explanation

To recognize the cell address, you can use the Name Box. The Name Box is located next to the Formula Bar in Microsoft Excel. It displays the cell reference or range name of the currently selected cell. By looking at the Name Box, you can easily identify the cell address of the selected cell.

23. A _____ is the intersection point of a column and a row.

- A. Column
 - B. Row
 - C. Cell
-

Correct Answer

C. Cell

Explanation

A cell is the intersection point of a column and a row. In a table or spreadsheet, columns are vertical and rows are horizontal. Each cell represents a unique intersection point where a specific piece of data can be entered or displayed. Therefore, a cell is the correct term to describe this intersection point.

24. Which sort of a program is the Microsoft Excel 2016?

- A. Word Processing Package.
 - B. Graphics Package
 - C. Spreadsheets Package
 - D. Multimedia Package
-

Correct Answer

C. Spreadsheets Package

Explanation

Microsoft Excel 2016 is a spreadsheets package. It is a powerful tool used for creating, organizing, and analyzing data in a tabular format. Excel offers various features such as formulas, functions, charts, and graphs that allow users to perform complex calculations and data analysis. It is widely used in businesses, finance, accounting, and many other fields for tasks like budgeting, data manipulation, and creating reports. Therefore, the correct answer is spreadsheets package.

25. Prepare the following worksheet:

Number of Schools in Selected Districts of Uganda

District	Secondary	Primary	Total
Kampala	68	114	
Jinja	124	186	
Iganga	78	64	
Mukono	39	84	
Wakiso	52	109	
Moroto			
Total			

After completion of the above worksheet do the following:

- Save the worksheet with name **school** by clicking on **Office Button** followed by **SAVE** or Clicking on to FLOPPY Icon. 
- Find out total number of schools in each district using formula i.e. **=b3+c3** (Remember formula starts with a = sign followed by cell reference and in between operator +, similar to what you do on calculator). You may copy this formula by dragging it to cells downward.
- Also calculate total number of secondary school and primary school using SUM function. (Guess why we are using function in place of formula =SUM(RANGE))
- Save worksheet and Exit from MS Excel (Hope you remember how to save and exit from MS Excel)
- Select cell containing column titles and align them to center by clicking on **Paragraph group** and clicking on **Centre Alignment text** and also make them **Bold**.
- Save above workbook and **exit from EXCEL** (Click on X on the top right corner)

26. **Open the worksheet created in the exercise -1 (school)** (Opening existing worksheet – Click on File Tab – Click on Open– click on **school**). Following will be displayed with Total column filled:

Number of Schools in Selected Cities of Rajasthan

District	Secondary	Primary	Total
Kampala	68	114	
Jinja	124	186	
Iganga	78	64	
Mukono	39	84	
Wakiso	52	109	
Moroto			

Total			
Average			

After completion of worksheet, do the following:

- Increase the column width of the column A to so as to accommodate complete text in column.
 - Increase column width of B, C and D together to accommodate data.
 - Find out average number of secondary school using function of average.
=AVERAGE(RANGE)
 - Copy this formula across to calculate average for primary and total number
 - Save the file again
 - Insert a column before District Click on Column Heading A – Click on Insert in Cells group
 - Add S.NO. as column title and enter numbers 1 to 5.
 - Add cells for minimum and maximum number of schools and calculate values these values. (Use =**MIN(RANGE)** and =**MAX(RANGE)**)
 - Insert a row between Jinja and Iganga Districts. Click onto Select row header of Mukono (entire row would be highlighted) and Choose **Home - Insert Sheet Row**
 - Add following data in the blank row
-
- Tororo 58 42
-
- Copy formula of total from Jinja to Tororo in column E
 - Observe that Total, Average and Minimum are automatically changed. Isn't it great again, no botheration about changing the formula or function?
 - Change the value of Secondary schools to **Jinja from 124 to 139** and see where the result changes.
 - Make Title of the table and heading row **BOLD** and Centre. Save the worksheet.
 - Save the worksheet (It is always advisable to keep saving worksheet by pressing CTRL S)
 - Try learning Merge and Centre
 - Add a column after Total with heading **% of Primary Schools to Total**.
 - Calculate above value (Hint: cell references of Primary Schools/ Total * 100)

t) Once % is calculated copy this to rest of the cells (down)

u) Save the worksheet and exit from EXCEL

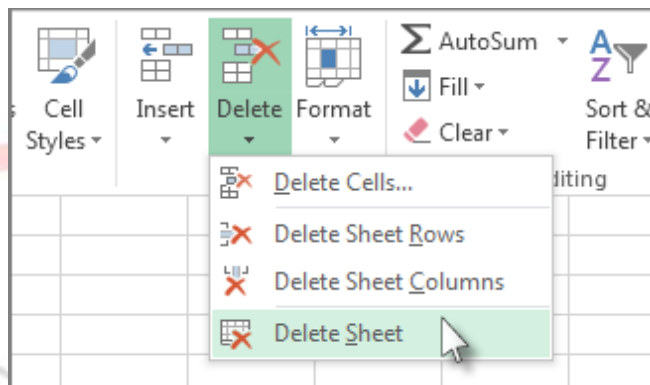
DAY #2

INSERTING AND DELETING WORKSHEETS

Once you've decided on a structure for your workbook, you may find that there are some worksheets that can be deleted.

Alternatively, you may find that you need additional blank worksheets inserted.

However, remember that deletion of worksheets is permanent and can't be undone using Undo, so always save your workbook before making these changes.



COPYING A WORKSHEET

Just as you can copy the contents of cells and ranges within a worksheet, you can duplicate worksheets within a workbook.

This technique is ideal for replicating layouts.

For example, if you have a budget workbook that contains data for several departments, you can create a worksheet for the first department and then copy it to create identical worksheets for other departments.

RENAMING A WORKSHEET

By default, Excel names worksheets as Sheet1, Sheet2, Sheet3, etc.

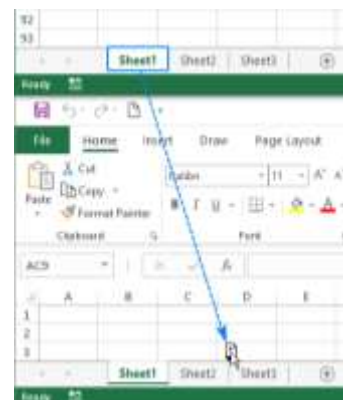
These names are fine if you are not planning to share the workbook, but changing these to something more relevant makes it much easier to understand the purpose of a worksheet.

You can also adjust the horizontal scroll bar to make room for longer, more meaningful worksheet names.

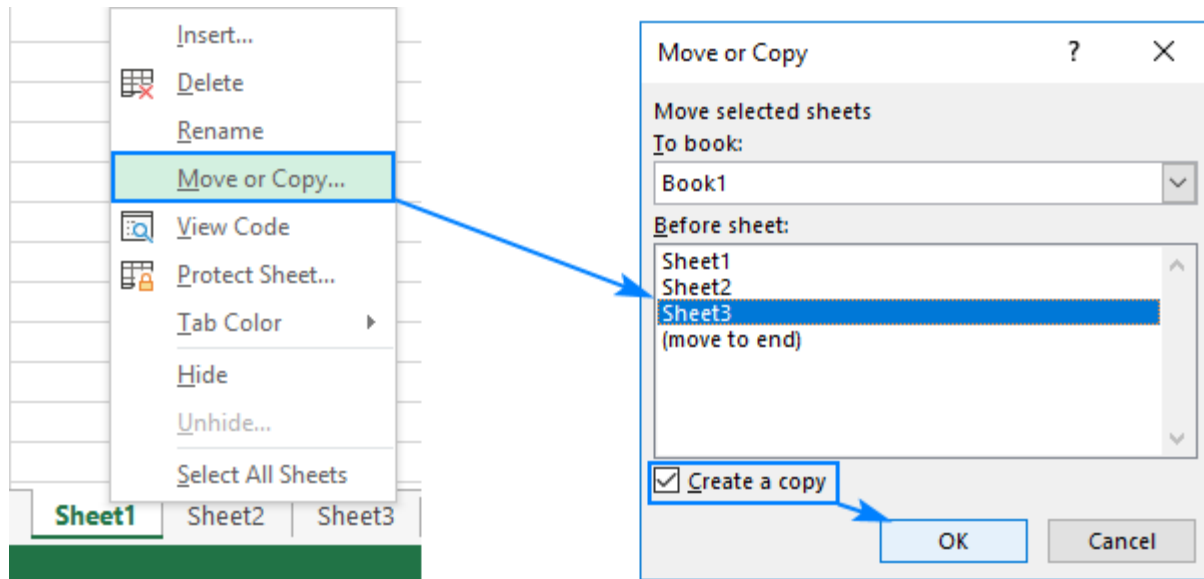
MOVING OR COPYING A SHEET TO ANOTHER WORKBOOK

You can copy worksheets to other workbooks as required.

For example, you might need to keep records for six different divisions – rather than send each division the entire set of records, you can copy their worksheet to another workbook and send them their data only.



If worksheets exist in the other workbook, you will need to determine the order in which to place the copied worksheet.

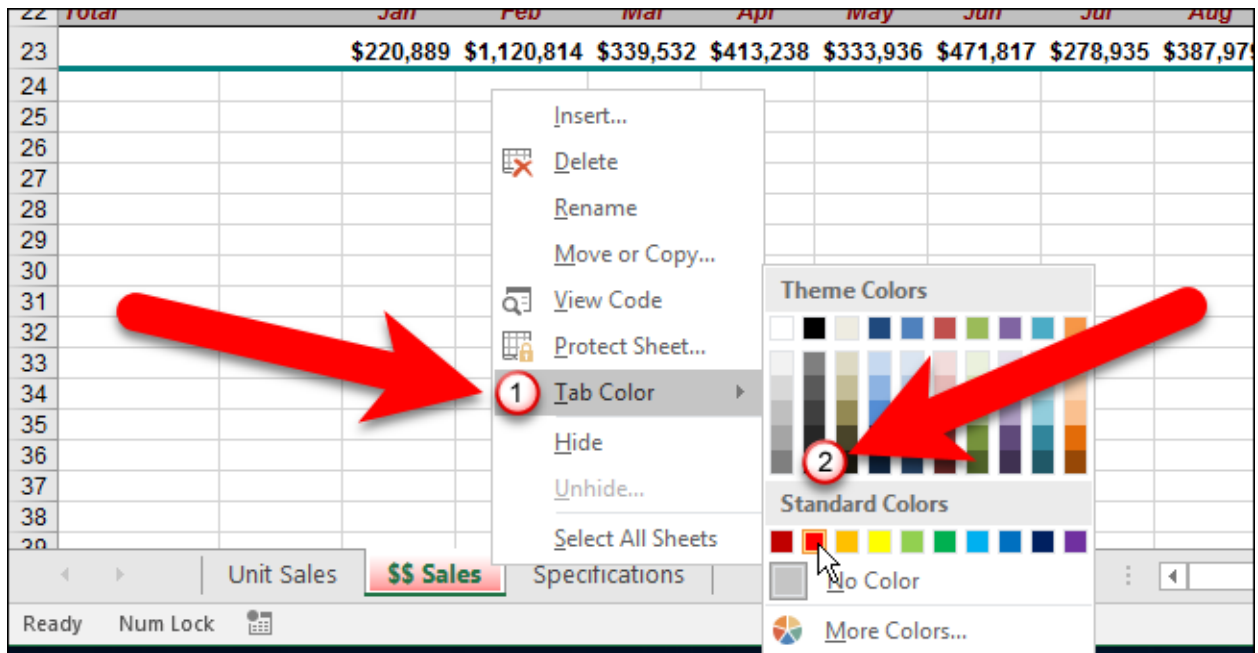


CHANGING WORKSHEET TAB COLOURS

To make it easier for you to distinguish between worksheets, Excel enables you to change the colors of worksheet tabs.

This allows you, for example, to quickly distinguish between different financial years, departments or months. The active sheet appears as underlined in a gradient version of the selected color, while inactive tabs will display a solid colour background.

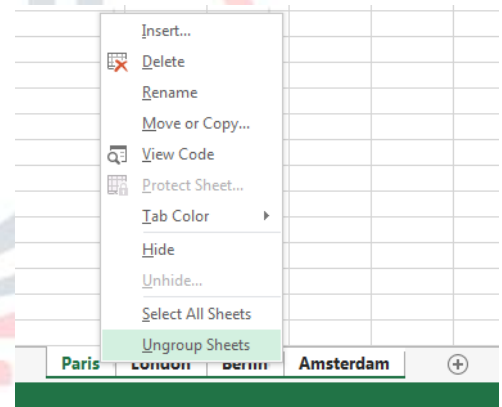




GROUPING WORKSHEETS

Worksheet grouping enables you to make the same change at once to all selected worksheets. This feature is useful in situations where your worksheets have identical layouts or text.

For example, if you want to format the heading for multiple worksheets, you simply group the worksheets, make a change to one worksheet and the other worksheets will reflect the change also.



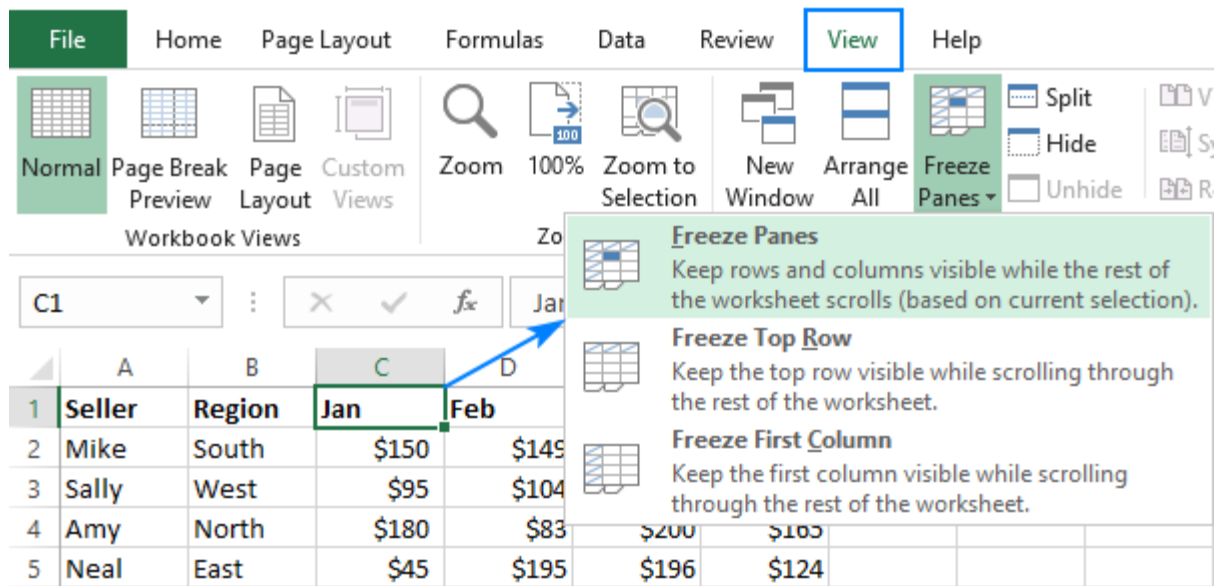
FREEZING ROWS AND COLUMNS

When you lay out your data in rows and columns, it is most likely that your headings end up at the top or to the left of your data.

If you have a large amount of data, you may find that when you scroll across or down to particular cells, the headings scroll out of view.

This problem can be resolved by freezing the rows and/or columns that hold the headings.

If you want to freeze only the rows above the selected cell (leaving all columns unfrozen), select the cell in column A of that row – e.g. to freeze rows 1 to 6, click in cell A7. The same applies to freezing only columns and leaving the rows unfrozen: select the cell in row 1.



EXERCISE

1. AN EXAMPLE OF CELL ADDRESS ...

- A. 23G
- B. 11
- C. A11
- D. A

Correct Answer

C. A11

Explanation

The given answer "A11" is an example of a cell address in a spreadsheet. In spreadsheets, cell addresses are used to identify and locate specific cells. In this case, "A11" represents a cell in column A and row 11. The letter "A" represents the column, and the number "11" represents the row. By using cell addresses, users can easily reference and manipulate data within a spreadsheet.

2. WHAT IS SHORTCUT FOR COPY ?

- A. CTRL+C
- B. CTRL
- C. COPY

D. PASTE

Correct Answer

A. CTRL+C

Explanation

The correct answer is CTRL+C. This is a commonly used keyboard shortcut for copying selected text or files. By pressing the CTRL key and the letter C at the same time, the selected content is copied to the clipboard, allowing it to be pasted elsewhere. This shortcut is widely recognized and used in various applications and operating systems.

3. HOW TO START ANY FORMULA IN A CELL ?

A. /

B. *

C. +

D. =

Correct Answer

D. =

Explanation

To start any formula in a cell, you need to use the "=" sign. This signifies to the spreadsheet software that you are entering a formula and not just typing in regular text. By using the "=" sign, you are instructing the software to perform calculations or functions based on the formula you input after the "=" sign.

4. WHICH SYNTAX IS CORRECT ?

A. =(A1-B1)

B. (A1-B1)

C. A1

Correct Answer

A. =(A1-B1)

Explanation

The correct syntax is =(A1-B1). This formula subtracts the value in cell B1 from the value in cell A1. The equal sign at the beginning indicates that it is a formula, and the parentheses are used to group the subtraction operation.

5. On an excel sheet the active cell is indicated by ?

- A. A dark wide border
 - B. WITH A DOT
 - C. COLOR OF THE CELL IS CHANGED
 - D. By italic text
-

Correct Answer

A. A dark wide border

Explanation

The active cell on an Excel sheet is indicated by a dark wide border. This border helps the user easily identify which cell they are currently working on. It stands out from the rest of the cells and provides a visual cue. This is a common feature in spreadsheet software to assist users in navigating and editing their data efficiently.

6. If you press __, the cell accepts your typing as its contents ?

- A. SHIFT+ENTER
 - B. ENTER
 - C. CTRL
 - D. CTRL+ENTER
-

Correct Answer

B. ENTER

Explanation

When you press "ENTER," the cell in question accepts your typing as its contents. This is because "ENTER" is the key that is typically used to confirm and finalize any input or changes made in a cell. By pressing "ENTER," you are indicating to the spreadsheet software that you have finished entering data or making edits, and it should accept and store the information you have inputted.

7. A EXCEL SHEET CONTAINS ?

- A. ROW
 - B. COLUMN
 - C. ROW AND COLUMN
-

D. LINES

Correct Answer**C. ROW AND COLUMN****Explanation**

An Excel sheet contains both rows and columns. Rows are horizontal lines that run across the sheet, while columns are vertical lines that run down the sheet. Rows are identified by numbers, while columns are identified by letters. The intersection of a row and a column is called a cell, and each cell can hold data such as text, numbers, or formulas. Therefore, an Excel sheet consists of both rows and columns to organize and store data effectively.

8. CAN WE ADD PICTURE IN A EXCEL SHEET ?

A. YES

B. NO

Correct Answer

A. YES

Explanation

Pictures can be added to an Excel sheet by using the "Insert" tab and selecting the "Picture" option. This allows users to insert images from their computer or from online sources into their Excel spreadsheet. Adding pictures can be useful for visualizing data or adding additional context to the information in the sheet.

9. Which of the following function, a formula with a logical function shows the word "TRUE" or "FALSE" as a result ?

A. IF

B. OR

C. AND

D. NOT

Correct Answer

A. IF

Explanation

The IF function is a logical function that evaluates a condition and returns "TRUE" or "FALSE" based on the result. It allows us to perform different actions based on whether the condition

is true or false. Therefore, the IF function is the correct answer as it shows the word "TRUE" or "FALSE" as a result.

10. PRODUCT FUNCTION IS USED FOR ?

- A. MULTIPLICATION
 - B. DIVISION
 - C. ADDITION
 - D. NOTHING
-

Correct Answer

A. MULTIPLICATION

Explanation

The product function is used for multiplication. It is a mathematical operation that combines two or more numbers to find their total value when they are multiplied together. The product function can be used to calculate the result of multiplying any given numbers, making it a useful tool in mathematical calculations and problem-solving.

11. What is Microsoft Excel?

- A. Microsoft excel is an electronic worksheet developed by Microsoft, to be used for organizing, storing and manipulating.
 - B. IT IS USED FOR VARIOUS CALCULATION
 - C. ANY OF THE ABOVE
-

Correct Answer

C. ANY OF THE ABOVE

Explanation

Microsoft Excel is a widely used electronic worksheet developed by Microsoft. It is primarily used for organizing, storing, and manipulating data. However, it is also extensively used for various calculations, making it a versatile tool for professionals in various fields. Therefore, the correct answer is "Any of the above," as Microsoft Excel can be used for organizing, storing, manipulating data, and performing calculations.

12. EVERY CELL MUST HAVE A ADDRESS ..

- A. TRUE
 - B. FALSE
-

Correct Answer

A. TRUE

Explanation

Every cell in a system or structure must have an address in order to be identified and located. This is essential for efficient organization and communication within the system. Without addresses, it would be difficult to refer to specific cells and track information accurately. Therefore, it is true that every cell must have an address.

13. TABLE OPTION CAN BE FOUND ON ...

- A. HOME
 - B. INSERT
 - C. FORMULAS
 - D. VIEW
-

Correct Answer

B. INSERT

Explanation

The correct answer is INSERT because the "Table" option is usually found in the "Insert" tab of various software applications, such as Microsoft Word or Excel. This tab contains various options for inserting different types of objects, including tables. Therefore, the correct option for finding the "Table" option is INSERT.

14. CTRL + X COMMAND IS USED FOR ?

- A. CUT
 - B. COPY
 - C. PASTE
-

Correct Answer

A. CUT

Explanation

CTRL + X command is used for cutting text or selected content from a document or file. This means that the selected content is removed from its original location and stored in the clipboard. It can then be pasted elsewhere using the CTRL + V command. This command is commonly used to move or transfer content within a document or between different documents or applications.

15. CTRL + Z IS USED FOR ?

- A. REDO
 - B. COPY
 - C. UNDO
 - D. PASTE
-

Correct Answer

C. UNDO

Explanation

The correct answer is UNDO because CTRL + Z is a keyboard shortcut commonly used to undo the previous action in many computer programs. This command allows users to reverse their last action and restore the document or file to its previous state. It is a convenient feature that helps users correct mistakes or revert changes they no longer want.

16. What is the symbol of deduction?

- A. +
 - B. '
 - C. *
 - D. None of these
-

Correct Answer

D. None of these

Explanation

The symbol of deduction is typically represented by the symbol "->" or "⊢" in logic and mathematics. However, none of the given options (+, ', *, None of these) represent the symbol of deduction. Therefore, the correct answer is "None of these."

17. The _____ function adds all of the indicated cells together and divides by the total number of cells.

- A. Best
 - B. Average
 - C. Worst
 - D. None of these
-

Correct Answer

B. Average

Explanation

The average function is used to calculate the average of a set of numbers. In this context, it adds all of the indicated cells together and then divides the sum by the total number of cells. Therefore, the correct answer for this question is "Average".

18. What is the symbol for division?

- A. -
- B. /
- C. +
- D. =

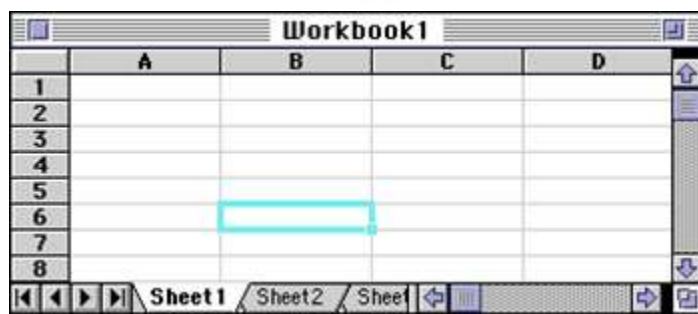
Correct Answer

B. /

Explanation

The symbol "/" is used to represent division in mathematical equations. It is commonly used to indicate the operation of dividing one number by another. For example, in the equation $10 / 2$, the symbol "/" is used to show that 10 is being divided by 2, resulting in a quotient of 5. Therefore, the correct answer is "/".

19. Name this cell:



- A. B2
- B. B5
- C. B6
- D. B8

Correct Answer

C. B6

Explanation

The correct answer is B6 because it is the only option that matches the given description. The question asks for the name of a cell, and B6 is the only cell mentioned in the list of options.

20. _____ are used to designate each row's location.

- A. Symbols
- B. Alphabets
- C. Signs
- D. Numbers

Correct Answer

D. Numbers

Explanation

Numbers are used to designate each row's location. In many systems, rows are assigned a number to indicate their position or order. This numbering system allows for easy identification and organization of rows within a dataset or table. By using numbers, it becomes clear which row is being referred to, making it more efficient to locate and work with specific rows of data.

21. Jolis Company Limited keeps the employee's payroll data in a worksheet as shown below:

Jolis Company Limited

S. No.	Name	Earning				Gross Salary
		Basic Pay	HRA	DA	CCA	
1	Abdi	178,000				
2	Abishag	123,500				
3	Abishai	27,500				
4	Asahel	69,500				
5	Olivia	48,000				
6	Lidya	30,500				
7	David	124,000				
Total						

- a) Enter the above data in a worksheet and save the workbook as **EMPLOYEES**.
- b) In the third column we need to type text BASIC PAY in the same cell and want that it should be displayed in two lines in the same cell.

- Select the entire row and click on  (**Wrap Text** in the **Alignment** Group of the **Home** Tab).
- c) Make Title of the Worksheet (**Jolis Company Limited**) as BOLD and also place in the Centre of the four Columns in which data is typed. (Select seven cells on top of the columns, click on to Merge and Centre Icon on the Tool Bar)
- d) Calculate HRA (16 % of Basic Pay), DA (21% of Basic Pay), CCA is fixed amount of UGX 1,000. Also calculate Gross Salary which is sum of Basic Pay, DA, HRA and CCA.
- e) Select the data you have entered and put border on all the cells using **Border in Font Group of HOME tab**. Save your workbook and exit.

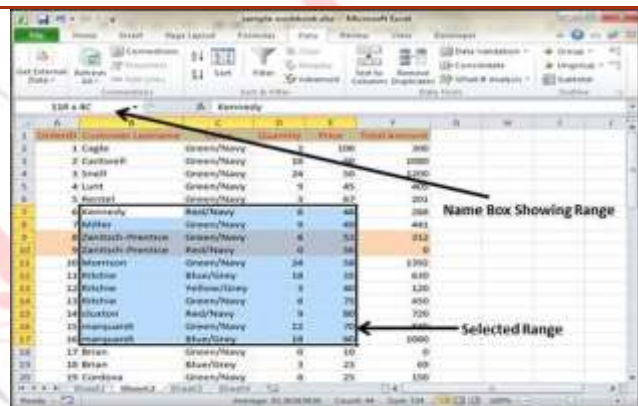
DAY #3

SELECTING RANGES

A **contiguous range** is any group of selected cells that form either a square or a rectangle. A single cell that is selected is also considered to be a range.

Ranges can be selected using the mouse, the keyboard or a combination of the two.

Once selected, you can use the range for input, or apply formatting, or copy the cells as required.

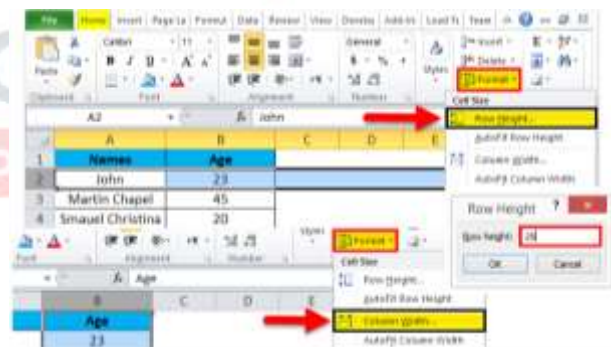


SELECTING ROWS

If you want to make changes to an entire row, such as bolding all of the headings in a row or changing the font of all the cell entries, you must first select the row.

This is done by clicking on the row header to the left of the row.

Remember that any changes you make will apply to every cell in the row all the way across to column XFD, so be careful!



SELECTING COLUMNS

If you want to make changes to an entire column, such as bolding all of the headings in a column or changing the font of all the cell entries, you must first select the column.

This is done by clicking on the column header directly above the column.

Remember that any changes you make will apply to every cell in the column all the way down to row 1,048,576!

	A	B	C	D	E	F	G	H	I
1	OrderDate	Region	Rep	Item	Units	Unit Cost	Total		
2	1/6/2018	East	Jones	Pencil	95	1.99	189.05		
3	1/23/2018	Central	Kivell	Binder	50	19.99	999.5		
4	2/9/2018	Central	Jardine	Pencil	36	4.99	179.64		
5	2/26/2018	Central	Gill	Pen	27	19.99	539.73		
6	3/15/2018	West	Sorvino	Pencil	56	2.99	167.44		
7	4/1/2018	East	Jones	Binder	60	4.99	299.4		

UNDERSTANDING FORMATTING

In Excel there are always two aspects to a number: how the number presents on the screen (known as formatting) and the underlying value of the number.

Take 2% as an example – on the screen it is formatted to appear as a number with a percentage sign, whereas the real value in the cell is .02.

Number Formatting – The Veil Placed Over Numbers

All calculations in Excel are performed using numbers – this is only logical.

So, when you want to perform a calculation, you type the numbers in various cells, then create formulas to reference those numbers.

How do you show what those numbers represent?

For example, how do you show you are working with currency, or percentages, or even dates (which in Excel are really numbers)?

Excel allows you to show these representations using number formatting. With number formatting you change the way a number looks so that it makes immediate sense to the reader of your worksheet. The underlying value of number, however, remains unchanged.

For example, instead of showing sales tax in a worksheet as .1 you show it as **10%**, to show 12889.95 as currency it would appear \$12,889.95 or €12,889.95 (depending upon the currency you are working with), and to show 44104 as a date you show it as **30-Sep-2020** (remember, dates are actually numbers representing the number of days from January 1, 1900).

The following worksheet contains formatted numbers:

	A	B	C	D	E	F	G	H	I
1	Sales Earnings								
2									
3	Employee			Date	Height	Weight		Com'n	
4	No	First Name	Last Name	Started	(Mtr)	(Kg)	Total Sales	%	Commission
5	2344	John	Smith	03-Oct-03	1 6/7	69.30	\$8,220,266.00	2%	164,405.32
6	3433	Mary	Henry	12-Apr-04	2 1/9	75.22	\$12,771,833.00	2%	255,436.66
7	3233	Harry	Ulin	02-Mar-99	1 4/5	87.90	\$35,324,399.00	2%	706,487.98
8	5445	Jim	Harrison	04-Jul-92	2 1/5	95.66	\$17,338,194.00	2%	346,763.88
9	3333	Larry	Graham	14-May-05	2	89.44	\$9,670,630.00	2%	193,412.60
10	4444	David	Jenkins	06-Feb-07	1 2/3	68.30	\$6,152,310.00	3%	184,569.30
11	3332	Ian	Quinn	26-Mar-95	1 6/7	69.32	\$36,973,644.00	3%	1,109,209.32
12	9887	Horace	Smyth	23-Dec-01	1 7/9	80.48	\$10,755,146.00	3%	322,654.38
13	4646	Yolanda	Victor	05-Jun-89	1 5/8	80.52	\$5,061,883.00	4%	202,475.32
14	5555	Quentin	Engels	03-Apr-01	1 8/9	78.40	\$13,329,586.00	5%	666,479.30

With the formatting removed from the numbers the worksheet looks as follows:

	A	B	C	D	E	F	G	H	I
1	Sales Earnings								
2									
3	Employee			Date	Height	Weight		Com'n	
4	No	First Name	Last Name	Started	(Mtr)	(Kg)	Total Sales	%	Commission
5	2344	John	Smith	37897	1.85	69.3	8220266	0.02	164405.32
6	3433	Mary	Henry	38089	2.1	75.22	12771833	0.02	255436.66
7	3233	Harry	Ulin	36221	1.797	87.9	35324399	0.02	706487.98
8	5445	Jim	Harrison	33789	2.21	95.66	17338194	0.02	346763.88
9	3333	Larry	Graham	38486	1.935	89.44	9670630	0.02	193412.6
10	4444	David	Jenkins	39119	1.65	68.3	6152310	0.03	184569.3
11	3332	Ian	Quinn	34784	1.862	69.32	36973644	0.03	1109209.32
12	9887	Horace	Smyth	37248	1.77	80.48	10755146	0.03	322654.38
13	4646	Yolanda	Victor	32664	1.62	80.52	5061883	0.04	202475.32
14	5555	Quentin	Engels	36984	1.9	78.4	13329586	0.05	666479.3

Formatting can also be applied as you type. For example, if you type **30/9/2020** Excel will place the number **44104** in the cell but will format this number as a date and show it as you typed it.

- In Microsoft Excel, dates are converted to serial numbers that represent the number of days that have passed since January 1, 1900. For example, the serial number for July 5, 2011 is 40,729.

There are also a range of number formatting options on the ribbon that allow you to apply formatting to numbers after they have been entered into a worksheet.

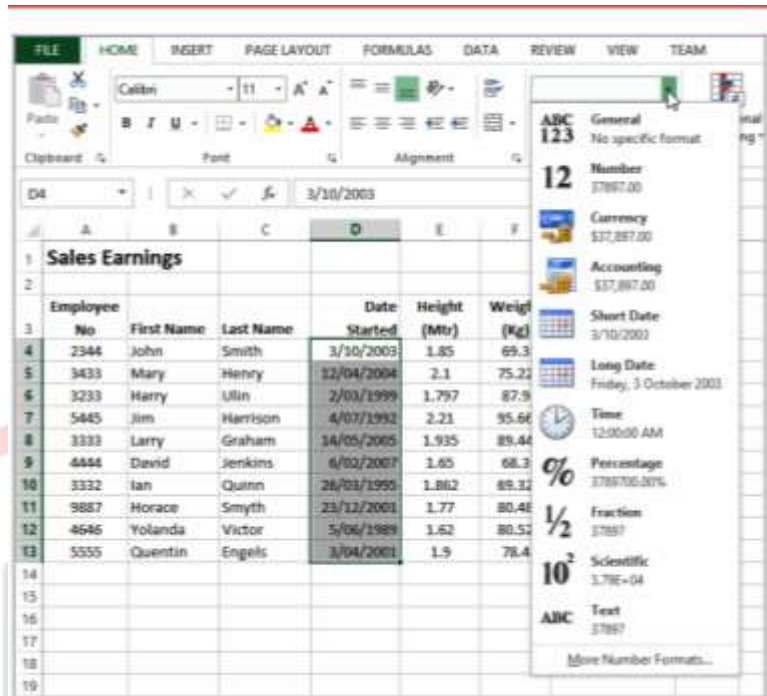
APPLYING GENERAL FORMATTING

The Number Format command in the Number group on the HOME tab contains a drop arrow that provides a gallery of the more commonly used number formats.

You can apply these formats easily and quickly to a selected cell or range of cells in the worksheet.

Excel may appear to round values up or down as necessary – however, the value in the cell does not change. Sometimes you'll see minor rounding discrepancies.

The Currency format shows the currency format and symbol appropriate to the country your computer is configured for.



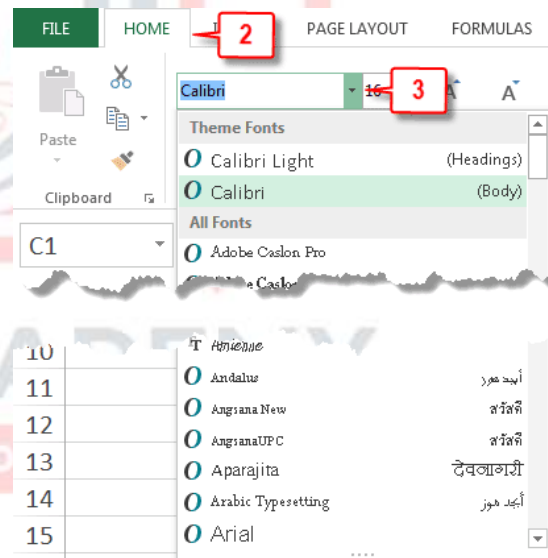
CHANGING FONTS

The appearance that you choose for your text is referred to as the font or typeface.

Font traditionally refers to a combination of typeface, style and size in points (e.g. Arial Bold 12 pt).

In Excel 2007, font just refers to the typeface or shape of the letters.

Typical classic fonts include Times New Roman, Arial, Century Gothic and Copperplate.



EXERCISE

1. The first cell in EXCEL worksheet is labeled as

- A. AA
- B. A1
- C. Aa
- D. A0

Correct Answer

B. A1

Explanation

The first cell in an Excel worksheet is labeled as A1. This is because Excel follows a specific naming convention for its cells, where the column letter is followed by the row number. In this case, the letter "A" represents the first column, and the number "1" represents the first row, hence the cell is labeled as A1.

2. What happens when dollar signs (\$) are entered in a cell address? (ex. \$B\$2:\$B\$10)

- A. An absolute cell address is created.
 - B. Cell address will change when it is copied to another cell.
 - C. The sheet tab is changed.
 - D. The status bar does not display the cell address.
-

Correct Answer

A. An absolute cell address is created.

Explanation

When dollar signs (\$) are entered in a cell address, an absolute cell address is created. This means that the cell reference will not change when it is copied to another cell. The dollar signs lock the column and row references, making it an absolute reference. This is useful when you want to refer to a specific cell or range of cells in a formula and you don't want the reference to change when copied to other cells.

3. What are the tabs that appear at the bottom of each workbook called?

- A. Reference tabs
 - B. Position tabs
 - C. Location tabs
 - D. Sheet tabs
-

Correct Answer

D. Sheet tabs

Explanation

The tabs that appear at the bottom of each workbook are called sheet tabs. These tabs allow users to navigate between different sheets within the workbook. Each sheet tab represents

a separate worksheet or a chart in the workbook. By clicking on a sheet tab, users can easily switch between different sheets and access the data or content stored in each sheet.

4. What is represented by the small, black square in the lower-right corner of an active cell or range?

- A. Copy handle
 - B. Fill handle
 - C. Insert handle
 - D. Border
-

Correct Answer

B. Fill handle

Explanation

The small, black square in the lower-right corner of an active cell or range is represented by the Fill handle. The Fill handle is used to quickly fill a series or pattern in adjacent cells. It can be dragged to copy the content of the active cell or range to other cells in a selected direction.

5. In Excel, a Data Series is defined as what?

- A. A type of chart
 - B. A cell reference
 - C. A collection of related data
 - D. A division of results
-

Correct Answer

C. A collection of related data

Explanation

A data series in Excel refers to a collection of related data. It is a set of values that are plotted in a chart or used in calculations. This allows for easy organization and analysis of data. A data series can be represented by a column, row, or range of cells in a worksheet. It helps to visually represent and understand the relationship between different data points.

6. B7:B9 indicates:

- A. Cells B7 and cell B9 only
 - B. Cells B7 through B9
-

- C. Cell B8 only
 - D. None of the above
-

Correct Answer

B. Cells B7 through B9

Explanation

B7:B9 indicates a range of cells starting from B7 and ending at B9, inclusive. This means it includes cells B7, B8, and B9.

7. The Cancel and Enter buttons appear in the:

- A. Title bar
 - B. Formula bar
 - C. Menu bar
 - D. Sheet tabs
-

Correct Answer

B. Formula bar

Explanation

The Cancel and Enter buttons appear in the Formula bar. The Formula bar is located at the top of the Excel window, just below the Ribbon. It displays the contents of the active cell and allows users to edit or enter formulas. The Cancel button is used to cancel any changes made in the Formula bar, while the Enter button is used to confirm and apply the changes.

8. MS-EXCEL can be used to automate

- A. Financial statements, Business forecasting
 - B. Transaction registers, inventory control
 - C. Accounts receivable, accounts payable
 - D. Any of the above
-

Correct Answer

D. Any of the above

Explanation

MS-EXCEL can be used to automate various tasks including financial statements, business forecasting, transaction registers, inventory control, accounts receivable, and accounts

payable. It is a versatile tool that allows users to create formulas, perform calculations, organize data, and generate reports. Therefore, it can be used to automate any of the mentioned tasks, making it a suitable answer.

9. NOT, AND, OR and XOR are

- A. Logical Operators
 - B. Arithmetic operators
 - C. Relational operators
 - D. None of the above
-

Correct Answer

A. Logical Operators

Explanation

The given correct answer is "Logical Operators" because NOT, AND, OR, and XOR are commonly used in computer programming as logical operators. These operators are used to manipulate boolean values and perform logical operations such as negation, conjunction, disjunction, and exclusive disjunction. They are essential for creating complex logical expressions and making decisions based on multiple conditions.

10. In a report, you need to show the monthly rainfall in Kampala. The best way to do this is to insert a

- A. Calendar
 - B. Photograph of rainfall
 - C. Chart showing rainfall amounts
 - D. Database of rainfall
-

Correct Answer

C. Chart showing rainfall amounts

Explanation

The best way to show the monthly rainfall in Nepal in a report would be to insert a chart showing rainfall amounts. This would provide a visual representation of the data, allowing the reader to easily understand and compare the rainfall amounts for different months. A chart would also allow for the inclusion of additional information such as average rainfall, trends, or any other relevant data points. This method is more effective than using a calendar or photograph, as it provides a clear and concise presentation of the information.

11. You want to record experiment information and create a chart that shows the rate of crystal growth over a period of time. The best application to use would be:

- A. Word processing
 - B. Spreadsheet
 - C. Database
 - D. Graphics
-

Correct Answer

B. Spreadsheet

Explanation

A spreadsheet would be the best application to use for recording experiment information and creating a chart that shows the rate of crystal growth over a period of time. Spreadsheets allow for easy data entry and organization, as well as the ability to perform calculations and create charts and graphs. This would enable the user to input the experiment data, calculate the rate of crystal growth, and create a visual representation of the data in the form of a chart.

12. You are editing a worksheet that you had previously saved. If you want to save the edited sheet without losing the original one, which command should you use?

- A. New
 - B. Save As
 - C. Edit
 - D. Save
-

Correct Answer

B. Save As

Explanation

To save the edited worksheet without losing the original one, the command "Save As" should be used. This command allows you to create a new copy of the worksheet with the changes made, while keeping the original version intact. By using "Save As," you can specify a new name or location for the edited sheet, ensuring that both versions are preserved.

13. If you want to have a blank line after the title in a worksheet, what is the best thing for you to do?

- A. Re-format the spreadsheet
 - B. Insert a row
-

- C. Increase the column width
 - D. Use the spacebar
-

Correct Answer

B. Insert a row

Explanation

To have a blank line after the title in a worksheet, the best thing to do is to insert a row. This will create a new row below the title, effectively giving a blank line for separation. Re-formatting the spreadsheet might involve more complex and unnecessary changes, increasing the column width would only affect the width and not create a blank line, and using the spacebar is not recommended as it can disrupt the formatting and alignment of the worksheet. Therefore, inserting a row is the most appropriate solution.

14. In order to arrange the countries from those with the highest population to those with the lowest, you need to sort on the population field in _____

- A. Ascending order
 - B. Descending order
 - C. Alphabetical order
 - D. Random order
-

Correct Answer

B. Descending order

Explanation

To arrange the countries from highest to lowest population, you need to sort them in descending order. This means that the countries with the highest population will be listed first, followed by those with lower populations. Sorting in ascending order would list the countries with the lowest population first, which is not what is required in this case. Alphabetical order would sort the countries based on their names, not their population. Random order would not provide any logical arrangement based on population.

15. In order to perform a calculation in a spreadsheet, you need to use a:

- A. Table
 - B. Formula
 - C. Field
 - D. Variable
-

Correct Answer

B. Formula

Explanation

To perform a calculation in a spreadsheet, you need to use a formula. A formula is a set of instructions that tells the spreadsheet how to calculate a specific result based on the values in the cells. It can include mathematical operations, cell references, and functions. By using formulas, you can automate calculations and update them easily when the input data changes.

16. A fast way to add up this column of numbers is to click in the cell below the numbers and then:

- A. Click Subtotals on the Data menu.
 - B. View the sum in the formula bar.
 - C. Click the AutoSum button on the Standard toolbar, then press ENTER.
-

Correct Answer

C. Click the AutoSum button on the Standard toolbar, then press ENTER.

Explanation

The fastest way to add up the column of numbers is to click the AutoSum button on the Standard toolbar and then press ENTER. This is because the AutoSum button automatically selects the range of cells above it and adds them up, saving time and effort. Pressing ENTER confirms the sum and displays the result in the selected cell. The other options mentioned, such as using Subtotals on the Data menu or viewing the sum in the formula bar, may also work but are not as efficient or straightforward as using the AutoSum button.

17. Say that you want to paste a formula result — but not the underlying formula — to another cell. You would copy the cell with the formula, then place the insertion point in the cell you want to copy to. What next?

- A. Click the Paste button on the Standard toolbar.
 - B. Click the arrow on the Paste button on the Standard toolbar, then click Formulas.
 - C. Click the arrow on the Paste button on the Standard toolbar, then click Values.
-

Correct Answer

C. Click the arrow on the Paste button on the Standard toolbar, then click Values.

Explanation

To paste a formula result without the underlying formula to another cell, you would first copy the cell with the formula. Then, you would place the insertion point in the cell where you want to paste the result. Next, you would click the arrow on the Paste button on the Standard toolbar, and finally, select the "Values" option. This will paste only the calculated value of the formula to the desired cell.

18. To add a new row, click a cell in the row immediately above where you want the new row.

- A. True
 - B. False
-

Correct Answer

B. False

Explanation

To add a new row in a table, you need to click on a cell in the row immediately below where you want the new row to be added, not above. Therefore, the statement "To add a new row, click a cell in the row immediately above where you want the new row" is false.

19. To copy an entire worksheet and all its data, you should click the worksheet tab of the sheet that you want to copy, hold down SHIFT, and then drag the selected sheet along the row of sheet tabs.

- A. True
 - B. False
-

Correct Answer

B. False

Explanation

To copy an entire worksheet and all its data, you should click the worksheet tab of the sheet that you want to copy, hold down CTRL, and then drag the selected sheet along the row of sheet tabs.

20. A user wishes to remove a spreadsheet from a workbook. Which is the correct sequence of events that will do this?

- A. Go to FILE - SAVE AS - SAVE AS TYPE - Excel 4.0 Work Sheet
 - B. Right click on the spreadsheet tab and select DELETE
 - C. Right click on the spreadsheet and select INSERT - ENTIRE COLUMN
-

Correct Answer

B. Right click on the spreadsheet tab and select DELETE

21. Click the _____ and select New to create a new, blank workbook.

- A. File Menu
 - B. Ribbon
 - C. Home tab
 - D. Microsoft Office Button
-

Correct Answer

A. File Menu

22. A _____ is a group of cells that form a rectangle on the screen.

- A. Label
 - B. Workbook
 - C. Worksheet
 - D. Column
 - E. Range
-

Correct Answer

E. Range

Explanation

A range is a group of cells that form a rectangle on the screen. It can consist of a single cell or multiple cells, and it is used to perform calculations, apply formatting, or manipulate data within the specified cells. A range can be selected, copied, moved, or formatted as a unit, making it a useful tool for managing and analyzing data in spreadsheets or other applications.

23. How do you know a range is active?

- A. It is highlighted on the screen.
 - B. It has a marquee around it.
 - C. The data changes to Bold to let me know it's active
 - D. I'm psychic & I just Know!!
-

Correct Answer

A. It is highlighted on the screen.

Explanation

The correct answer is "It is highlighted on the screen." This suggests that the range is visually emphasized or stands out in some way on the screen, making it easy to identify as active.

24. An example of a range address.

- A. A1:B3
 - B. A1;B3
 - C. A1-B3
 - D. A1=B3
 - E. A1*B3
-

Correct Answer

A. A1:B3

Explanation

The correct answer is A1:B3 because it represents a range of cells in a spreadsheet. The "A1" refers to the top-left cell of the range, and the "B3" refers to the bottom-right cell of the range. This notation is commonly used in spreadsheet programs to specify a group of cells that you want to work with or apply a formula to.

25. _____ are used to designate each column's location.

- A. Letters
 - B. Numbers
 - C. Symbols
 - D. Images
-

Correct Answer

A. Letters

Explanation

Letters are used to designate each column's location. In most cases, columns in a document or spreadsheet are labeled with letters, starting from A and going up to Z. This labeling system allows for easy reference and organization of data within the columns. By using

letters, it becomes simple to identify and locate specific columns when working with large amounts of information.

26. In a spreadsheet, the _____ is defined as the vertical space that is going across the window.

- A. Symbol
 - B. Column
 - C. Row
 - D. None of these
-

Correct Answer

B. Column

Explanation

In a spreadsheet, the column is defined as the vertical space that is going across the window. It is used to organize and categorize data in a tabular format. Each column is identified by a letter at the top of the spreadsheet, such as A, B, C, etc. Columns are used to input, calculate, and display data in a structured manner, making it easier to analyze and manipulate the information within the spreadsheet.

27. In a spreadsheet, the _____ is defined as a space where a specified row and column intersect.

- A. Labels
 - B. Cell
 - C. Formulas
 - D. Numbers
-

Correct Answer

B. Cell

Explanation

A cell in a spreadsheet is defined as a space where a specified row and column intersect. It is the basic unit of a spreadsheet where data can be entered and manipulated. Cells can contain various types of information such as text, numbers, formulas, and labels. They allow for organization and calculation of data within the spreadsheet.

28. _____ are entries that have equations that calculates the value to display.

- A. Labels
-

- B. Constants
- C. Spreadsheet
- D. Formulas

Correct Answer

D. Formulas

Explanation

Formulas are entries that have equations that calculate the value to display. They are used in various applications, such as spreadsheets, to perform calculations and manipulate data. Formulas can include mathematical operations, functions, and references to other cells or ranges. By using formulas, users can automate calculations and update values dynamically based on changes in input data.

29. Open the file EMPLOYEE created in Exercise 3 and add following columns after Gross Salary

Deductions		Net Salary
GPF	SI	Total

a) Calculate GPF and SI, rules for calculating these are as follows.

- i. GPF - 6000 is deducted if basic pay is less than 55000, else 8500 is deducted.
- ii. SI, which is 2000 for everyone irrespective of grade pay. Also fill up Total Deduction and Net Salary.

b) Save the workbook again.

c) Change Basic Pay of two or employee and check whether the effects of above changes are reflected in gross salary and net salary.

d) Save the file.

e) Change the formula for calculating SI premium in such a way that 20% of basic pay is deducted for employees getting basic pay more than 80000 and for others SI deduction is 15%.

30. Prepare the following worksheet:

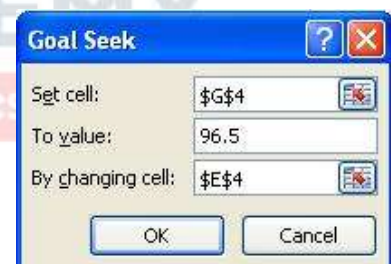
ABC Consultants, Jaipur

Item	Quantity	Unit Price	Gross Value
I7 Computer System	2	55000.00	
Notebook i5	2	43000.00	
Printer	2	12000.00	

Pen Drive	15	560.00	
Total Gross			
Discount		7%	
Amount Payable			
Tax on Amount Payable		4.5%	
Net Amount Payable			

- Increase the width of the Column A in such a way that entire data is accommodated in the column.
- Enter a formula to calculate the gross value for I7 Computer System. (**Remember * is used for Multiplication**)
- Copy this formula for rest of the items
- Calculate the total of the Gross Values for the all the products using SUM Function.
- Calculate the Discount, which has been given as 7%. (Do not panic if you type 7% and it shows .07).
- Calculate the Amount Payable after subtracting Discount Amount form the Total Gross and also calculate Tax and Net Amount Payable.
- Change the discount rate to 8% and see the changes. Also change Tax Rate to 6.25% and observe the changes.
- Make Title of the Worksheet (AMC Consultants, Jaipur) as BOLD and also place in the Centre of the four Columns in which data is typed. (Select four cells on top of the columns, Click on to Merge and Centre Icon on the Font Group of the HOME tab.
- Change discount percentage value from 7% to 10% and see what changes are reflected. Save the worksheet as ITEM.
- Using Goal Seek** - Suppose you want that you have an amount available with you of UGX169800. than how much discount should be offered to you so that you are able to purchase items indicated above. How will you approach this issue - GUESS.

Select **DATA tab – Data Tools – What if Analysis - GOAL SEEK** option and specify desired information

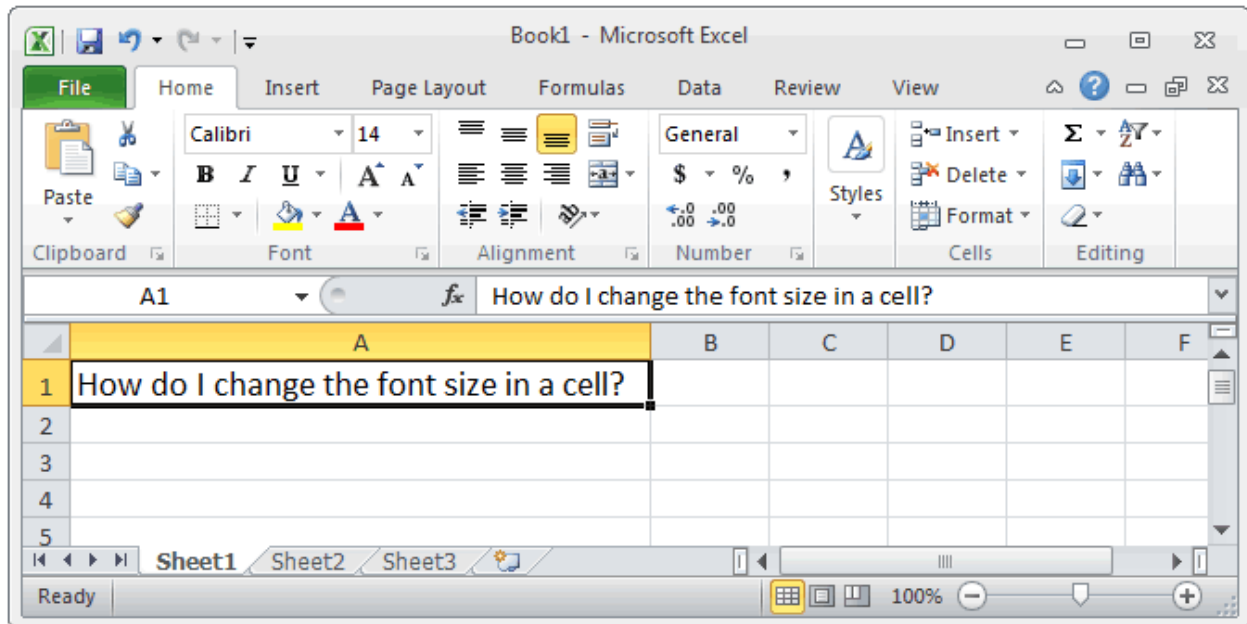


DAY #4

CHANGING FONT SIZE

One way that text can be emphasized is by changing the size of the font. For example, if your normal text is 11 pt., you may like to make the headings 13 pt. or larger. Font size may also be changed for small detailed items; such as comments or a caption.

Main headings in a worksheet usually appear in a slightly larger font size compared to the rest of the data.



UNDERSTANDING BORDERS

Borders are lines that are placed around the edges of individual cells or ranges. The lines may be thin, thick, solid, dashed, black or colored, or even double lines.

The reason for using borders is that the lines can be used to group together data or indicate totals, or to draw the user's attention to critical cells that may need special data entry.

Here are some examples.

A Worksheet without and with Borders

Borders can be used to apply a structure.

Here's the same worksheet shown without borders and then with borders applied. The use of borders helps to highlight the totals and separate them from the other data.

Alpheius Global Enterprises
Budget Forecast for Year

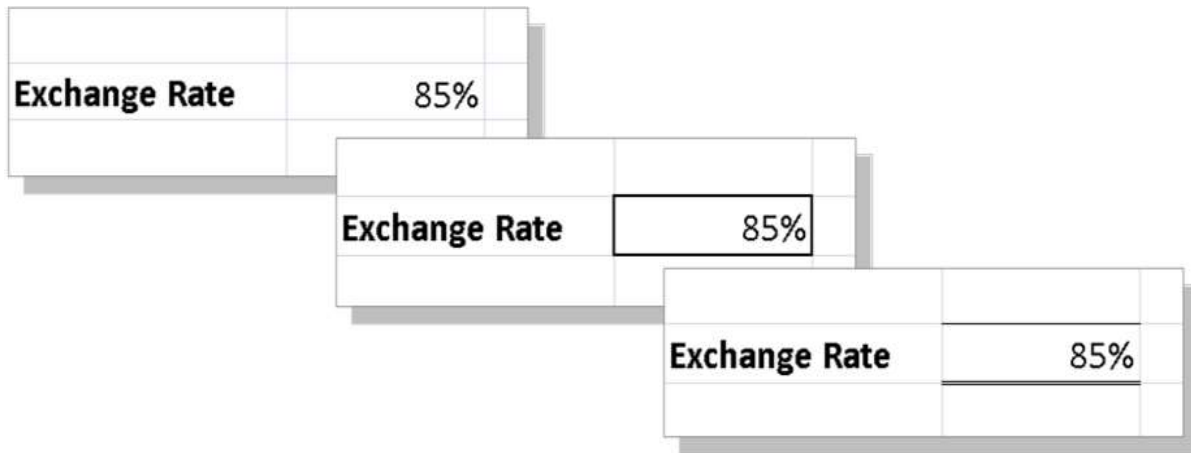
Sales	Jan	Feb	Mar	Apr	May	Jun	Total \$AUS	Total \$US
Auckland	\$105,025	\$154,700	\$140,837	\$163,721	\$100,093	\$198,102	\$950,477	\$665,334
Dublin	\$152,429	\$168,555	\$159,385	\$175,964	\$193,582	\$212,941	\$1,063,476	\$744,434
Melbourne	\$252,149	\$298,545	\$274,122	\$301,534	\$331,688	\$364,857	\$1,922,894	\$1,346,026
New York	\$253,123	\$262,189	\$245,408	\$268,940	\$296,934	\$326,627	\$1,654,212	\$1,157,949
Total Sales	\$863,726	\$983,989	\$820,344	\$911,179	\$1,002,297	\$1,102,526	\$5,581,060	\$3,913,742

Costs	Jan	Feb	Mar	Apr	May	Jun	Total \$AUS	Total \$US
Auckland	\$55,100	\$85,055	\$81,887	\$90,076	\$99,084	\$108,992	\$520,195	\$364,136
Dublin	\$83,822	\$92,678	\$87,911	\$96,703	\$106,573	\$117,810	\$584,497	\$409,140
Melbourne	\$193,688	\$264,135	\$250,777	\$265,895	\$282,441	\$300,685	\$1,057,602	\$740,321
New York	\$139,267	\$144,145	\$134,955	\$140,451	\$163,296	\$179,625	\$698,739	\$496,817
Total Costs	\$471,877	\$486,033	\$455,531	\$501,085	\$551,199	\$606,112	\$2,072,831	\$1,450,422
Gross Income	\$390,849	\$497,955	\$372,813	\$410,094	\$451,104	\$496,214	\$2,519,329	\$1,763,320
Fixed Costs	\$2,000	\$2,200	\$2,420	\$2,662	\$2,928	\$3,221	\$15,431	\$10,802
Net Income	\$388,849	\$495,755	\$370,393	\$407,432	\$448,176	\$492,993	\$2,503,898	\$1,752,518
Exchange Rate	95%							

➤ Border Variations

Borders can be applied to all four sides of a cell, or to individual sides of a cell.

The following examples show a cell without a border, with an outside border and a top and double bottom border.



APPLYING A BORDER TO A RANGE

You can apply a border to a range of cells. This allows you to place an outline around them to indicate that the cells are somehow related to each other, or to place borders between cells to indicate that they are in separate groups.

Borders can be used in ranges of cells to create a more form-like appearance. The borders available for single cells can also be applied to ranges.

	A	B	C	D	E	F	G	H	I
1	Alpheius Global Enterprises								
2	Budget Forecast for Year								
3									
4									
5	Sales	Jan	Feb	Mar	Apr	May	Jun	Total \$AUS	Total \$US
6	Auckland	\$105,025	\$154,700	\$148,837	\$163,721	\$180,093	\$198,102	\$950,477	\$807,906
7	Dublin	\$152,429	\$168,555	\$159,985	\$175,984	\$193,582	\$212,941	\$1,063,476	\$903,955
8	Melbourne	\$352,149	\$298,545	\$274,122	\$301,534	\$331,688	\$364,857	\$1,922,894	\$1,634,460
9	New York	\$253,123	\$262,189	\$245,400	\$269,940	\$296,934	\$326,627	\$1,654,212	\$1,406,080
10									
11	Total Sales	\$862,726	\$883,989	\$828,344	\$911,179	\$1,002,297	\$1,102,526	\$5,591,060	\$4,752,401
12									
13	Costs	Jan	Feb	Mar	Apr	May	Jun	Total \$AUS	Total \$US
14	Auckland	\$55,100	\$85,055	\$81,887	\$90,076	\$99,084	\$108,992	\$520,195	\$442,165
15	Dublin	\$83,822	\$92,678	\$87,911	\$96,703	\$106,373	\$117,010	\$584,497	\$496,822
16	Melbourne	\$193,688	\$164,155	\$150,777	\$165,855	\$182,441	\$200,685	\$1,057,602	\$898,961
17	New York	\$139,267	\$144,145	\$134,955	\$148,451	\$163,296	\$179,625	\$909,738	\$773,278
18									
19	Total Costs	\$471,877	\$486,033	\$455,531	\$501,085	\$551,193	\$606,312	\$3,072,031	\$2,611,227
20									

WRAPPING AND MERGING TEXT

Microsoft Excel will allow long cell entries to spill across to other adjacent cells to the right as long as those cells are empty.

If those cells contain data, the spill-over will be chopped off. If you need to place long text entries in a cell you can arrange for Microsoft Excel to wrap the text within the cell and also merge that cell with others to accommodate the longer text entry.

The screenshot shows the Microsoft Excel interface. The 'Home' tab is selected in the ribbon. In the 'Alignment' group, the 'Wrap Text' button is highlighted with a red box. A red arrow points from this button to a table in the worksheet. The table is located in the range A1 to J9 and is outlined with a red border. The table contains the following data:

	A	B	C	D	E	F	G	H	I	J
1	Order ID	Date	Customer Name	Product	Category	Quantity	Price	Total	Region	
2	1001	1/15/2024	John Smith	Laptop	Electronics	2	750	1500	North	
3	1002	1/18/2024	Jane Doe	Smartphone	Electronics	5	400	2000	East	
4	1003	1/20/2024	Bob Johnson	Tablet	Electronics	3	300	900	South	
5	1004	2/1/2024	Alice Brown	Office Chair	Furniture	7	120	840	West	
6	1005	2/10/2024	Chris Green	Desk	Furniture	4	200	800	North	
7	1006	2/15/2024	John Smith	Monitor	Electronics	2	150	300	North	
8	1007	3/3/2024	Jane Doe	Desk Lamp	Furniture	10	30	300	East	
9	1008	3/8/2024	Bob Johnson	Headphones	Electronics	6	50	300	South	
10										
11										

EXERCISE

1. After you merge a range into a single cell, you can't realign its content.

- A. True
- B. False

Correct Answer

B. False

Explanation

When you merge a range into a single cell, you can still realign its content. Merging cells is a formatting feature in which multiple adjacent cells are combined into a single cell. This is commonly used to create headings or to merge cells with similar data. After merging, you

can still adjust the alignment of the content within the merged cell, such as aligning it to the left, center, or right. Therefore, the statement that you can't realign the content after merging is false.

2. Once you hide a row, it remains hidden as long as the workbook is active.

- A. True
 - B. False
-

Correct Answer

A. True

Explanation

When you hide a row in a workbook, it will remain hidden as long as the workbook is active. This means that even if you close the workbook and reopen it, the row will still be hidden. This feature allows you to temporarily hide certain rows of data without permanently deleting them. To unhide the row, you would need to activate the workbook and manually unhide the row.

3. All functions have arguments. There are no functions which work without providing them arguments.

- A. True
 - B. False
-

Correct Answer

B. False

Explanation

This statement is false because there are functions called "void functions" that do not require any arguments. Void functions are used when a function does not need any input parameters to perform its task. They can be called without providing any arguments. Therefore, not all functions require arguments.

4. Although 3D Charts are visually attractive, they can obscure the relationship between the values in the chart by making it difficult to see when slice is larger.

- A. True
 - B. False
-

Correct Answer

A. True

Explanation

3D charts can indeed be visually attractive, but they can also make it difficult to accurately interpret the relationship between the values in the chart. This is because the depth and perspective of the chart can obscure the size of each slice or bar, making it challenging to determine which slice or bar is larger or smaller. As a result, the relationship between the values may not be accurately conveyed, leading to potential misinterpretation of the data. Therefore, the statement is true.

5. Excel has a built-in format to display 4-digit year values, so you do not need to create one.

A. True

B. False

Correct Answer

A. True

Explanation

Excel does have a built-in format to display 4-digit year values. This means that users do not need to create a custom format for displaying 4-digit year values in Excel. Therefore, the statement "True" is correct.

6. Formatting data only change the appearance of data, it does not affect the data itself.

A. True

B. False

Correct Answer

A. True

Explanation

Formatting data refers to changing the visual representation of data without altering the underlying values or calculations. This can include changing the font, color, alignment, or adding borders to cells. The purpose of formatting is to enhance the readability and presentation of data. However, formatting does not modify the actual data or its values. Therefore, the statement that formatting only changes the appearance of data and does not affect the data itself is true.

7. If a formula contains several functions, Excel starts with the outermost function and then moves inward.

A. True

B. False

Correct Answer

B. False

Explanation

Excel actually starts with the innermost function and then moves outward when evaluating a formula that contains several functions. This means that Excel will first calculate the innermost function and then use the result of that calculation as an argument for the next outer function, and so on, until it reaches the outermost function. Therefore, the correct answer is False.

8. An easy way to sort data when there is only one sort field is to use the Sort A to Z or Sort Z to A buttons.

A. True

B. False

Correct Answer

A. True

Explanation

When there is only one sort field, using the Sort A to Z or Sort Z to A buttons is indeed an easy way to sort data. These buttons allow the data to be sorted in ascending (A to Z) or descending (Z to A) order based on the selected field.

9. You can create maximum one Excel table in a worksheet

A. True

B. False

Correct Answer

B. False

Explanation

The statement is false because you can create multiple Excel tables in a worksheet. Excel allows users to create and manage multiple tables within a single worksheet. Each table can have its own set of columns, rows, and formatting options. This feature is useful for organizing and analyzing data in different ways within the same worksheet.

10. As you begin to type a function name within a formula into a cell, a list of functions that begin with the letters you typed appears.

A. True

B. False

Correct Answer

A. True

Explanation

When you start typing a function name in a cell, a list of functions that start with the letters you have typed appears. This feature is called "AutoComplete" and it helps users by suggesting relevant function names as they type, saving time and reducing errors. Therefore, the given statement is true.

11. Prepare a worksheet as under and fill in the blanks using Worksheet features.

Abishag Financial Corporation

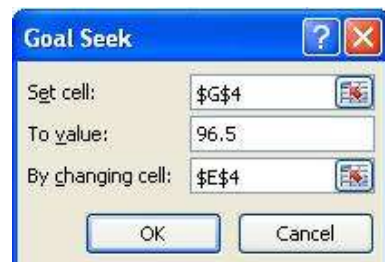
S.No	Particulars	Target	Expenditure / Utilized			% Utilization
			Up to last month	Current month	Total	
1.	Civil Works at the head Office	345000	115000	145000		
2.	Gardening	80000	7000	15000		
3.	Machinery & Equipment for the office	315000	273000	17000		
4.	Vehicles	277000	0	217000		
5.	Training for officers and employees	70000	42300	0		
6.	Technical Assistance	5000	1000	3000		
7.	Furniture/Fixture	15000	2000	9000		
8.	Salaries & Allowances	490430	139000	47000		
9.	Operating Costs	71000	13000	21000		
10.	Miscellaneous	30000	1000	2500		
	Total					

- a) Format the worksheet as above. Most of the features to format the worksheet have already been covered in the previous exercises. Select Column heading and give them different colors. (Home – Styles - Cell Styles and select appropriate Style).

b) Calculate % utilization of the target. Select different formatting for cells for % Utilization

c) Save the worksheet as RFC.

d) **Using Goal Seek** - Suppose you want that % utilization for Civil Works should be 96.5%, then how much amount should be the expenditure in the current month. How will you approach this issue?



12. The following statement shows that actual expenditure incurred on various account heads at JPesa International for last three years. Prepare the following worksheet.

Budget of JPesa International

S. No.	Account Head	Actual Expenditure			Average of last three years	Estimate for 2020-21	% increase over 2017-18
		2017-18	2018-19	2019-20			
1	Salaries	800000	830000	850000			
2	Travel Expenses	70000	90000	100000			
3	Medical Charges	60000	70000	75000			
4	Office Expenses	110000	130000	170000			
5	Advt., Sales & Publicity Expenses	8000	9000	12000			
	Total :						

After completion of worksheet, perform the following:

- Place Title (Budget of JPesa International) in the center of the table.
- Format the heading row of the table in such a way that data is displayed as given above, i.e. longer text is displayed in more than one line in the same cell. (Hint: Select the cells and click  on (**Wrap Text** in the **Alignment** Group of the **Home** Tab).
- Make heading row as **BOLD** and **CENTER**
- Calculate the total expenditure for all the account head for each year. Also calculate average expenditure for last three years.
- Calculate the estimates for the year 2020-21 by assuming that it is 15% higher than the average of last three years
- Calculate the increase in percentage for the year 2020-21 with respect to the year 2017-18. (Increase is calculated after subtracting 2017-18 expenses from the estimate for the year 2020-21 and dividing it by expenditure of the year 2017-18.

- g) Draw Border around the Table by Clicking onto **BORDER** Icon on the Font Group of **HOME** tab.
- h) Save the worksheet as **BUDGET**.
- i) Exit form MS Excel.

DAY #5

UNDERSTANDING FUNCTIONS

Imagine having to create a formula that calculated the monthly payments on a loan, or the average of over 100 cells – these would require complex or long formulas that would be time consuming to develop. This is the role of hundreds of arithmetic functions that have been pre-programmed in Excel for you.

Functions Overview

Functions are simply pre-programmed formulas already provided for you in Excel which can perform calculations covering a wide range of categories including statistics, date and time arithmetic, financial calculations, lists, engineering, and more.

Just like normal formulas that you create, functions must start with an equal sign. The equal sign is then followed by the name of the function (usually a descriptive name which indicates the purpose of the function).

Most functions also require additional information known as arguments which are supplied to the function in brackets after the function name. Functions are therefore written as follows:

=name(arguments)

The arguments are quite often cell or range references that contain values that can be used in the function.

For example, the commonest function is the **SUM** function which, as its name suggests, is used to sum or add values together. If you wanted to add all of the values in the cells from **B10** to **D15** you would write this function as:

=SUM (B10:D15)

As you can see this is much simpler than writing your own referential formula which would look like:

=B10+B11+B12+B13+B14+B15+D10+D11+D12+D13+D14+D15

Imagine writing and proofing a formula where you had to add 200 cells!

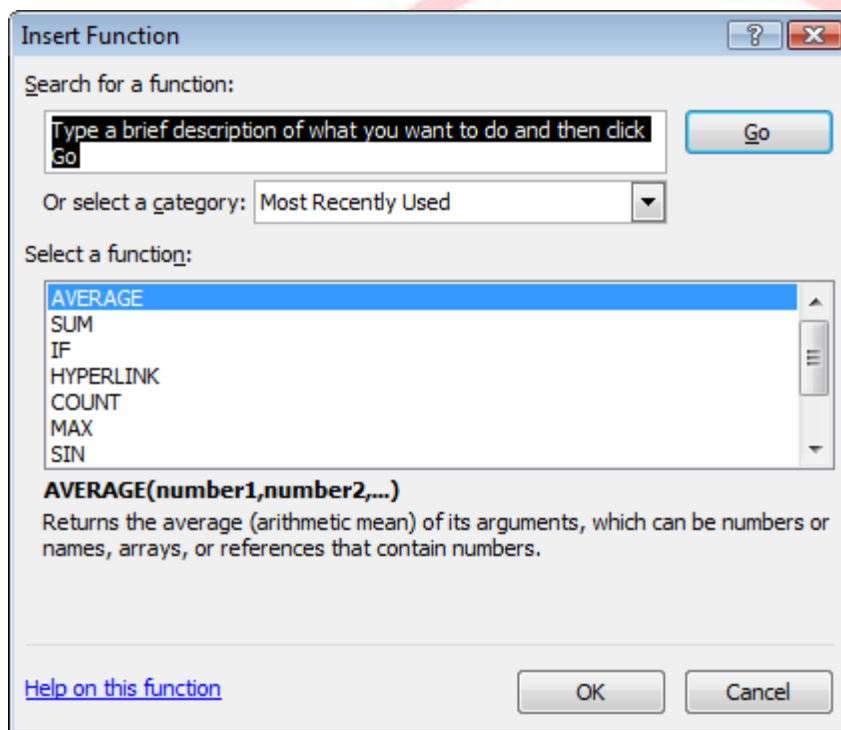
Typing Functions

If you are familiar with the function that you need you can type it into a cell exactly the same way you type any other formula.

If you are not sure if Excel has a function or you can't quite remember how it is written you can use the **Insert Function** tool on the Formula Bar to assist you.

When you click on this tool the **Insert Function** dialog box will be presented to you which lists the most recently used or common functions and also allows you to search for other functions that you might need.

The **Insert Function** dialog box will also type the function out for you and then provide you with a further dialog box to guide you through the process of specifying the arguments that the function needs to perform its calculation.



USING THE SUM FUNCTION TO ADD

One of the most used functions is the **SUM** function. This function allows you to add the values in a range of cells. The function is written as:

=SUM(range or ranges to add)

You can type the function, and then use the pointing technique to fill in the arguments. Excel then paints marquees around the cells involved helping you to track your progress.

F2							
	A	B	C	D	E	F	G
1	Subject Name	John	Ron	Harry	Jenifer	Total	
2	Maths	89	98	67	99	353	
3	English	90	87	76	90	343	
4	Science	67	65	89	67	288	
5	Economics	65	54	54	98	271	
6	Accounts	90	65	87	78	320	
7							

CALCULATING AN AVERAGE

The **AVERAGE** function allows you to average the values in a range of cells. It is written in much the same way as the SUM function, for example,

=AVERAGE (range of cells to average).

The average function can be applied using the Functions Wizard, a part of Excel that steps you through the process of creating a function or you can type it in yourself if you are comfortable with it.

AVERAGE											

FINDING A MINIMUM VALUE

The **Minimum** or **MIN** function allows you to extract the lowest value from a range of values. It is written in much the same way as the **SUM** function. For example,

=MIN(range of cells).

The function can be applied using the Function Wizard, or by typing the function in detail directly into the cell.

	A	B	C	D	E
1	Fruit	Qty in stock	MIN		
2	Apple	366	Case 1		
3	Mango	476	354	=MIN(366,476,398,982,354,534,408)	
4	Orange	398	Case 2		
5	Pineapple	982	354	=MIN(B2,B3,B4,B5,B6,B7,B8)	
6	Grape	354	Case 3		
7	Melon	534	354	=MIN(B2:B8)	
8	Banana	408			
9					
10					
11					

LOOKUP

The LOOKUP function in Excel retrieves a value from a one-row or one-column range. It performs a rough match lookup either vertically or horizontally. The vector form syntax is

=LOOKUP (lookup_value, lookup_vector, [result_vector]),

While the array form is

=LOOKUP (lookup_value, array).

lookup_value lookup_vector result_vector
=LOOKUP(B2,D2:D5,E2:E5)

	A	B	C	D	E
1		Result		Seller	Product
2	Seller	Sally		Adam	Bananas
3	Product	Oranges		Robert	Apricots
4				Sally	Oranges
5				Tom	Apples

HLOOKUP & VLOOKUP

B6 : **=HLOOKUP(B5, B1:J3, 2, FALSE)**

	A	B	C	D	E	F	G	H	I
1	Planet	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
2	Diameter	4,878	12,104	12,755	6,790	142,796	120,660	51,118	49,528
3	Temperature, °F	468	869	59	-9.4	-238	-292	-391	-391
4									
5	Planet	Earth							
6	Diameter	12,755							

HLOOKUP stands for Horizontal Lookup and can be used to retrieve information from a table by searching a row for the matching data and outputting from the corresponding column.

While **VLOOKUP** searches for the value in a column, HLOOKUP searches for the value in a row.

EXERCISE

1. Reference to the above image of Command Group "Alignment" which of the following tabs consist it:
-



- A. View
 - B. Formatting
 - C. Home
-

Correct Answer

C. Home

Explanation

The Command Group "Alignment" consists of the tabs: View, Formatting, and Home.

2. You can have results of any database table into Excel Worksheet, and by one click you can refresh the results?
-

- A. True
 - B. False
-

Correct Answer

A. True

Explanation

This statement is true because Excel has the capability to connect to various database systems and retrieve data from tables. Once the connection is established and the data is imported into an Excel worksheet, you can set up a refresh option that allows you to update the data with just one click. This ensures that the results in the worksheet are always up to date with the database table.

3. When using a database, changing the value of one cell in the range will not affect the bar size of all the other cells.
-

- A. True
 - B. False
-

Correct Answer

B. False

Explanation

When using a database, changing the value of one cell in the range will affect the barsize of all the other cells. This is because databars are designed to represent the relative values of the cells in the range, so when one value changes, the barsize of all other cells will adjust accordingly to maintain the relative proportions. Therefore, the correct answer is False.

4. The Insert Function dialog box tells you how to use functions.

A. True

B. False

Correct Answer

A. True

Explanation

The statement is true because the Insert Function dialog box is a feature in many software applications, such as Microsoft Excel, that provides a user-friendly interface for selecting and inserting functions into a cell or formula. It typically includes a search bar, a list of available functions, and a description or explanation of how to use each function. This dialog box serves as a helpful tool for users who may not be familiar with all the available functions or how to properly use them.

5. When a worksheet is printed, the grid lines that surround the cells are printable by default.

A. True

B. False

Correct Answer

B. False

Explanation

By default, when a worksheet is printed, the grid lines that surround the cells are not printable. To print the grid lines, the user needs to enable the "Print Gridlines" option in the print settings. Therefore, the statement "When a worksheet is printed, the grid lines that surround the cells are printable by default" is false.

6. When you clear a filter, all filters in the worksheet are cleared, if worksheet contains more than one filters

A. True

B. False

Correct Answer

B. False

Explanation

When you clear a filter, only the specific filter you cleared will be cleared, not all filters in the worksheet. This means that if the worksheet contains multiple filters, the other filters will remain applied and unchanged.

7. The function arguments dialog box tells you how to use functions.

A. True

B. False

Correct Answer

A. True

Explanation

The function arguments dialog box provides information on how to use functions. It likely includes details on the input parameters that a function requires, as well as any optional arguments and their respective data types. This information helps users understand the correct way to use functions in their code. Therefore, the statement "The function arguments dialog box tells you how to use functions" is true.

8. The simplest and most convenient way to add a record to a table is to enter the data in the first blank row.

A. True

B. False

Correct Answer

A. True

Explanation

Entering data in the first blank row is the simplest and most convenient way to add a record to a table because it ensures that the new record is added at the end of the existing data. This method avoids any potential conflicts or errors that may arise when trying to insert a record in a specific location within the table. Additionally, it allows for easy organization and readability of the table, as new records are appended to the bottom.

9. If you insert a new row at the top of the table, the new row becomes the header row and is formatted with the table style.

A. True

B. False

Correct Answer

B. False

Explanation

When you insert a new row at the top of the table, the existing header row remains unchanged and does not become the new row. The new row that is inserted will not automatically be formatted with the table style. Therefore, the correct answer is False.

10. In Excel, charts do not remain linked or connected to the data sources if they appear in different worksheets.

A. True

B. False

Correct Answer

B. False

Explanation

In Excel, charts can remain linked or connected to the data sources even if they appear in different worksheets. This means that any changes made to the data in the source worksheet will be automatically reflected in the chart, regardless of which worksheet the chart is located in. Therefore, the statement that charts do not remain linked or connected to the data sources if they appear in different worksheets is false.

11. You can rotate a 3D chart in only two directions: horizontally along the x-axis and vertically along the y-axis.

A. True

B. False

Correct Answer

A. True

Explanation

The statement is true because in a 3D chart, you can rotate it horizontally along the x-axis, which means you can change the angle of view from left to right or vice versa. Additionally, you can also rotate it vertically along the y-axis, allowing you to change the angle of view from top to bottom or vice versa. These two directions of rotation give you the ability to explore different perspectives and angles of the 3D chart.

12. Pressing ENTER moves the selection one cell to the right by default.

- A. True
 - B. False
-

Correct Answer

B. False

Explanation

Pressing ENTER moves the selection one cell down by default, not one cell to the right.

13. Ctrl+B key makes the shortcut for making font in selected cell bold.

- A. True
 - B. False
-

Correct Answer

A. True

Explanation

The statement is true because pressing the Ctrl+B key combination in a selected cell will make the font in that cell bold. This is a common shortcut used in many software programs, including spreadsheet applications like Microsoft Excel.

14. What key should be used to edit the content of a cell?

- A. F1
 - B. F2
 - C. F3
-

Correct Answer

B. F2

Explanation

The F2 key should be used to edit the content of a cell. This key allows the user to directly edit the text or formula within a cell without having to go into the formula bar or double-clicking on the cell. By pressing F2, the cursor is placed at the end of the cell's content, allowing for easy editing and modification.

15. What function offers you the possibility to view different results depending on the entered condition?

- A. =If()
 - B. =Sum()
 - C. =SumIf()
-

Correct Answer

A. =If()

Explanation

The =If() function offers you the possibility to view different results depending on the entered condition. With this function, you can set up a logical test and specify what should happen if the test is true or false. This allows you to create conditional statements and display different results based on the conditions you specify.

16. A user wishes to remove a worksheet from a workbook. Which is the correct sequence of events that will do this?

- A. Go to file - save as - save as type - excel 4.0 worksheet
 - B. Right click on the worksheet tab and select delete
 - C. Right click on the worksheet and select insert - entire column
-

Correct Answer

B. Right click on the worksheet tab and select delete

Explanation

The correct sequence of events to remove a worksheet from a workbook is to right-click on the worksheet tab and select delete. This action will delete the selected worksheet from the workbook.

17. What is the keyboard shortcut for creating a chart from the selected cells?

- A. F3
 - B. F7
 - C. F9
 - D. F11
-

Correct Answer

D. F11

Explanation

Pressing the F11 key on the keyboard creates a chart from the selected cells. This shortcut allows users to quickly generate a chart without having to go through the menu options or use the mouse. By selecting the desired cells and pressing F11, users can easily visualize their data in a chart format, making it easier to analyze and interpret the information.

18. Using the autosum button will place in the selected cell ____.

- A. The sum of values in the cell's column
 - B. Nothing until you selects a range of cells
 - C. The sum of the cell's row unless you change the range
 - D. A formula which will add values in the range Excel guesses you want to add
-

Correct Answer

D. A formula which will add values in the range Excel guesses you want to add

Explanation

When you use the autosum button, Excel will automatically generate a formula in the selected cell that adds up the values in the range it guesses you want to add. This saves you the time and effort of manually typing out the formula.

19. The view that puts a blue line around each page that would be printed is the ____.

- A. Print Preview
 - B. Normal
 - C. Page Break Preview
 - D. Split View
-

Correct Answer

C. Page Break Preview

Explanation

The correct answer is "Page Break Preview". This view is used in programs like Microsoft Excel to show how the content of a document will be divided and printed on separate pages. It helps users to identify where page breaks will occur and adjust the layout accordingly. The blue line around each page in this view indicates the boundaries of each printed page.

20. The cell reference for a range of cells that starts in cell B1 and goes over to column G and down to row 10 is ____.

- A. B1-G10
-

- B. B1.G10
 - C. B1;G10
 - D. B1:G10
-

Correct Answer

D. B1:G10

Explanation

The correct answer is B1:G10. This is because the colon (:) is used to indicate a range of cells, and B1:G10 represents a range that starts from cell B1 and goes over to column G and down to row 10.

21. If you press ____, the cell accepts your typing as its contents but the focus stays in the same cell.

- A. ENTER
 - B. CTRL + ENTER
 - C. TAB
 - D. INSERT
-

Correct Answer

B. CTRL + ENTER

Explanation

Pressing CTRL + ENTER allows you to input your typing into the cell, but the focus remains in the same cell. This means that after typing, you can continue editing or formatting the cell without having to navigate to a different cell.

22. Once data is entered into a cell, can it be edited after saving the file?

- A. Yes, but it has to be saved as a different file name.
 - B. No, once it is saved it cannot be changed.
 - C. Yes, even after saving the file, double click on the cell where data is to be changed to edit
-

Correct Answer

C. Yes, even after saving the file, double click on the cell where data is to be changed to edit

Explanation

The correct answer states that data entered into a cell can be edited after saving the file. This can be done by double-clicking on the cell where the data is located, allowing for changes to be made.

23. To select several cells or ranges that are not touching each other, you would ____ while selecting

- A. Hold down the CTRL key
 - B. Hold down the SHIFT key
 - C. Hold down the ALT key
-

Correct Answer

- A. Hold down the CTRL key

Explanation

To select several cells or ranges that are not touching each other, you would hold down the CTRL key while selecting. Holding down the CTRL key allows you to select multiple non-adjacent cells or ranges by clicking on each of them individually. This is a useful feature when you need to select specific cells or ranges scattered throughout a spreadsheet without selecting everything in between.

24. What is a Workbook?

- A. A table of data that is organized into rows and columns
 - B. The Excel file that holds your worksheets
 - C. A built-in formula in Excel
-

Correct Answer

- B. The Excel file that holds your worksheets

Explanation

A workbook in Excel refers to the file that contains multiple worksheets. It is the main file where you can create, edit, and organize your worksheets. Each worksheet within the workbook can contain different data, formulas, and formatting. Therefore, the correct answer is "The Excel file that holds your worksheets."

25. A Function is?

- A. A symbol that represents a specific action
 - B. A way of arranging data in a particular order
 - C. A built-in formula in Excel
-

Correct Answer

C. A built-in formula in Excel

Explanation

A function in Excel is a built-in formula that performs a specific action or calculation on data. It is used to manipulate and analyze data in a spreadsheet. Functions can be used to perform mathematical operations, manipulate text, calculate dates and times, and much more. They are predefined formulas that can be easily applied to cells or ranges in a worksheet to automate calculations and save time.

26. A Range in Microsoft Excel refers to...

- A. A group of cells
 - B. A calculation that describes the relationship between cells
 - C. A way of arranging data in a particular order
-

Correct Answer

A. A group of cells

Explanation

A range in Microsoft Excel refers to a group of cells. This means that multiple cells can be selected and manipulated together as a single unit. Ranges are commonly used for performing calculations, applying formatting, or entering data across multiple cells simultaneously. By selecting a range, users can easily perform operations on a specific set of cells, making data management and analysis more efficient.

27. Which is the part of the Excel screen where you can find all the commands or buttons, depending on the tab you selected before?

- A. Ribbon
 - B. Formula Bar
 - C. Tab
-

Correct Answer

A. Ribbon

Explanation

The ribbon in Excel is the part of the screen where you can find all the commands or buttons, depending on the tab you selected before. It is located at the top of the Excel window and is divided into tabs, such as Home, Insert, Page Layout, and more. Each tab contains specific commands and options related to different tasks and functions in Excel. The ribbon provides

easy access to various tools and features, allowing users to navigate and perform actions efficiently.

28. What is the function of the autofit ?

- A. A symbol that represents a specific action
 - B. A way of arranging data in a particular order
 - C. Automatically change the width of a column to fit the longest entry or change the height of a row to fit the font
-

Correct Answer

C. Automatically change the width of a column to fit the longest entry or change the height of a row to fit the font

Explanation

The function of the autofit is to automatically adjust the width of a column to fit the longest entry or change the height of a row to fit the font. This feature is useful in spreadsheet programs or table editors where the content of cells may vary in length or font size. Autofit ensures that all the content is visible within the cell without any truncation or overlapping.

29. A worksheet is..

- A. A symbol that represents a specific action
 - B. Refers to the change a formula makes when it is moved or copied to other cells with different data
 - C. A table of data that is organized into rows and columns
-

Correct Answer

C. A table of data that is organized into rows and columns

Explanation

The correct answer is a table of data that is organized into rows and columns. A worksheet in the context of computer software, such as Microsoft Excel, refers to a document or file that contains a grid-like structure where data can be entered, organized, and manipulated. This grid is typically divided into rows and columns, forming a table-like structure. Users can input data into individual cells, perform calculations using formulas, and analyze and present the data in various ways. Worksheets are commonly used for tasks such as budgeting, data analysis, and creating charts or graphs.

30. What is a Chart?

- A. It is a visual way to display and compare data
-

- B. A way of arranging data in a particular order
 - C. A calculation that describes the relationship between cells
-

Correct Answer

A. It is a visual way to display and compare data

Explanation

A chart is a visual representation that allows for the display and comparison of data. It provides a clear and concise way to present information in a graphical format, making it easier to understand and analyze trends, patterns, and relationships within the data. By using different types of charts, such as bar graphs, pie charts, or line graphs, data can be visually organized and interpreted, enabling users to make informed decisions based on the information presented.

31. If you have data that runs across five columns, from column A to column E, one way to center a title across the top of the data would be to:

- A. Type the data in A1 and Merge cells A1:A5
 - B. Type the data in A1 and Merge cells A1:E1
 - C. Type the data in C1
 - D. Choose Center from the Format menu
-

Correct Answer

B. Type the data in A1 and Merge cells A1:E1

Explanation

To center a title across the top of the data that runs across five columns, the correct way would be to type the data in cell A1 and then merge cells A1 to E1. This will create a single merged cell spanning across all five columns, allowing the title to be centered across the top of the data.

32. In the following formula: =IF(A1="YES","DONE","RESTART"), what happens if A1="NO"?

- A. Formula returns the value "DONE"
 - B. Formula will show an error
 - C. Formula returns the value "RESTART"
 - D. Formula returns the value "YES"
-

Correct Answer

C. Formula returns the value "RESTART"

Explanation

The formula =IF(A1="YES","DONE","RESTART") checks the value in cell A1. If the value is "YES", it returns "DONE". If the value is anything other than "YES", including "NO", it returns "RESTART". Therefore, if A1="NO", the formula will return the value "RESTART".

33. After Excel displays an existing entry in the column that matches the characters you type, press the ____ key to accept the entry.

- A. Shift
 - B. Tab
 - C. Esc
 - D. Alt
-

Correct Answer

B. Tab

Explanation

When Excel displays an existing entry in the column that matches the characters typed, pressing the Tab key is the correct action to accept the entry. The Tab key allows the user to move to the next cell in the spreadsheet, effectively accepting the displayed entry and advancing to the next input.

34. When creating an Excel workbook, you should avoid using ____ text and background together, since this is the most common form of color blindness.

- A. Red and green
 - B. Yellow and blue
 - C. Black and white
 - D. Red and yellow
-

Correct Answer

A. Red and green

Explanation

When creating an Excel workbook, it is important to avoid using red and green text and background together. This is because red-green color blindness is the most common form of color blindness, and individuals with this condition may have difficulty distinguishing

between these colors. Using red and green together can make it challenging for color-blind users to read and interpret the information in the workbook accurately.

35. The _____ allows you to use Excel to change one or more values in a spreadsheet and then assess the effect those changes have on the calculated values.

- A. What-if analysis
 - B. Track Changes
-

Correct Answer

A. What-if analysis

Explanation

What-if analysis allows you to use Excel to change one or more values in a spreadsheet and then assess the effect those changes have on the calculated values. This feature is useful for exploring different scenarios and understanding how changes in input variables can impact the outcome. It helps in making informed decisions and analyzing the sensitivity of the model to different inputs. Track Changes, on the other hand, is a different feature in Excel that allows you to track and review changes made to a workbook by different users.

36. To sort data in an Excel worksheet you can use the ____.

- A. Sort A to Z and Sort Z to A buttons
 - B. Quicksort Options
 - C. Sort command on the Formulas tab
 - D. Any of the above
-

Correct Answer

D. Any of the above

Explanation

The correct answer is "Any of the above." This is because all of the options mentioned can be used to sort data in an Excel worksheet. The Sort A to Z and Sort Z to A buttons are available on the Home tab and can be used to sort data alphabetically in ascending or descending order. The Quicksort Options provide additional sorting options and customization. The Sort command on the Formulas tab can be used to sort data based on formulas or calculations. Therefore, any of these options can be used to sort data in Excel.

37. =FUNCTION(argument1,argument2, ...) Is _____.

- A. The syntax of writing optional arguments.
- B. An example of a formula.

- C. The general syntax of all functions.
 - D. The general syntax of all formulas.
-

Correct Answer

C. The general syntax of all functions.

Explanation

This answer is correct because the given statement "Is ____" is asking for the type of information that can be filled in the blank. The statement "The general syntax of all functions" accurately describes what can be expected when using the =FUNCTION() formula in any context. It suggests that the function can take multiple arguments, and the syntax for writing these arguments is the same for all functions.

38. If you have columns that are truncating your content, but you cannot increase the width of the worksheet and be able to print on one page, a solution for making the text visible without losing data would be to:

- A. Abbreviate all text
 - B. Reduce the font size to 8 pt
 - C. Click Wrap text
 - D. Delete one column
-

Correct Answer

C. Click Wrap text

Explanation

Clicking the "Wrap text" option allows the text to be displayed on multiple lines within a single cell, instead of being truncated. This ensures that all the content is visible without losing any data.

39. Labels are aligned at the ____ edge of the cell.

- A. Left
 - B. Right
 - C. Top
 - D. Bottom
-

Correct Answer

A. Left

Explanation

Labels are aligned at the left edge of the cell. This means that the text of the labels will be positioned towards the left side of the cell, while the remaining space in the cell will be on the right side of the label.

40. You can complete a cell entry by pressing:

- A. Enter
 - B. Tab
 - C. Shift+tab
 - D. An arrow key
 - E. Another cell
 - F. None of above
-

Correct Answer

A. Enter

Explanation

Pressing ctrl+enter or enter completes a cell entry in Microsoft Excel. This keyboard shortcut allows the user to stay in the current cell after entering data, rather than moving to the next cell. This can be useful when the user wants to quickly enter data in multiple cells without navigating to different cells each time.

41. The default font keyed in a new workbook is:

- A. 12 pt. Times New Romans
 - B. 11 pt. Calibri
 - C. 12 pt. Bradley Hand ITC
 - D. 10 pt. Times New Roman
-

Correct Answer

B. 11 pt. Calibri

Explanation

The default font keyed in a new workbook is 11 pt. Calibri.

42. If you choose a larger font size, the height of the row is automatically made taller.

- A. True
-

B. False

Correct Answer

A. True

Explanation

When a larger font size is chosen, the height of the row is automatically adjusted to accommodate the larger text. This is because larger text requires more vertical space to be properly displayed. Therefore, the statement "If you choose a larger font size, the height of the row is automatically made taller" is true.

43. Which button allows you to copy cell formats from one cell to another?

- A. Enter
 - B. Autofit
 - C. Format painter
 - D. Esc
-

Correct Answer

C. Format painter

Explanation

The format painter button allows you to copy cell formats from one cell to another. This feature is useful when you want to apply the same formatting, such as font style, color, and borders, to multiple cells without having to manually format each one. By selecting the cell with the desired format and then clicking on the format painter button, you can easily apply the same format to other cells in just a few clicks.

44. A _____ is a group of cells that form a rectangle on the screen.

- A. Label
 - B. Workbook
 - C. Worksheet
 - D. Column
 - E. Range
-

Correct Answer

E. Range

Explanation

A range is a group of cells that form a rectangle on the screen. It can include a single cell or multiple cells, and is used to perform calculations, apply formatting, or manipulate data within the specified cells. A range can be selected and modified as a unit, making it a convenient way to work with data in a structured manner.

45. Which Excel feature is used to organize data in alphabetical or numerical order?

- A. Sort
 - B. Filter
 - C. PivotTable
 - D. Conditional Formatting
-

Correct Answer

A. Sort

Explanation

The correct answer is "Sort". The Sort feature in Excel is used to organize data in alphabetical or numerical order. It allows users to arrange data in ascending or descending order based on a selected column. This feature is helpful for quickly finding and analyzing specific information in a large dataset.

46. Which function is used to find the average of a range of numbers in Excel?

- A. AVERAGE
 - B. SUM
 - C. COUNT
 - D. MAX
-

Correct Answer

A. AVERAGE

Explanation

The AVERAGE function is used to find the average of a range of numbers in Excel. It calculates the sum of the numbers in the range and then divides that sum by the count of numbers in the range. This function is commonly used to calculate the average value of a set of data, such as grades, sales figures, or test scores.

47. When you start to select a range of cells, you will see a thick white, cross-shaped pointer called a _____.

- A. Cell selector
- B. Cell cross
- C. Selection pointer
- D. Range selector

Correct Answer

C. Selection pointer

Explanation

When you start to select a range of cells, you will see a thick white, cross-shaped pointer called a selection pointer. This pointer allows you to indicate the specific cells or range of cells that you want to work with in a spreadsheet or other software program. It helps to visually highlight the cells being selected and assists in accurately defining the desired range for various operations such as formatting, copying, or deleting data.

48. Prepare following spreadsheet to see the expenses incurred on various items.

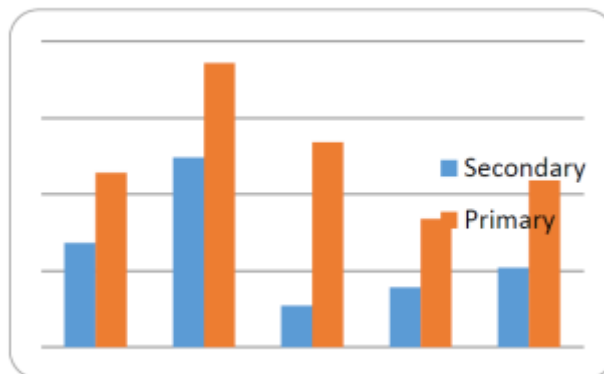
S. No.	Item Name	Amount Spent	% Expenses of Total Expenditure
1	House Rent	30000	
2	Milk	9700	
3	Kitchen	17500	
4	Bills	19500	
5	Education	12000	
6	Petrol	25000	
7	Entertainment	5600	
8	Miscellaneous	8000	
	Total		
	Salary Received	235000	
	Savings from Salary		

- a) Format the worksheet as given above and also calculate total expenditure.
- b) Salary Received is also given; therefore, calculate monthly savings from the salary.
- c) Also calculate % expenses incurred on each head of the total. Remember when you copy the formula down; the cell references are automatically changed. Therefore, you need to keep total as constant by using in the formula before row number of the total.
- d) How much would you be saving if Entertainment expenses were increased to 1350?
- e) Save the worksheet as **HOUSE-BUDGET**. Exit from MS Excel.

49. Open the worksheet **school** created on **Day #1 (Exercise 1 #7)** which is as follows.

Prepare a COLUMN chart-taking District on the X-axis and number of Secondary Schools on the Y-axis using the following steps:

- Select data consisting of **district names, secondary school and primary school**.
- Click onto INSERT TAB and select **COLUMN** from **CHARTS** group.
- Select **Column** chart type.
- The chart as displayed will be shown.



DAY #6

COMMON ERROR MESSAGES

Microsoft Excel has some in-built messages that can assist you when something goes wrong with a formula. These messages appear in the cell that contains the formula, and sometimes also other formula cells that depend upon it.

The messages are always prefixed with a hash sign (#) and appear with a code. The more common error messages are listed below.

➤ A Line of Hash (#) Signs

B2		:	X	✓	<i>fx</i>	540000234778
	A	B	C	D	E	F
1						
2		#####				
3						
4						
5						

Sometimes referred to as “tramlines”, a line of hash signs usually occurs because a column is not wide enough to display the numbers in the cell or formula.

Widening the column will correct this problem – you can drag the column heading until the value in the cell appears as it should.

➤ #DIV/0!

This message means you are trying to divide a value by zero – this is mathematically impossible.

In the example at the left we are trying to find the average number of persons per household.

All is fine as long as there is a value greater than zero in cell B3 (Houses).

B5 : ✕ ✓ <i>f_x</i> =B2/B3				
	A	B	C	D
1				
2	People	192,664		
3	Houses	0		
4				
5	Persons/house	#DIV/0!		
6				

As soon as we change this to a zero an error message appears in the formula cell (B5).

To prevent the error, you will need to enter a value greater than zero into cell B3, the divisor cell.

➤ #VALUE!

In this message Excel is advising that something in the formula is not a value and therefore a calculation can't be made.

A close examination of the example at the left shows cell B3 contains the word "three".

Therefore, the formula in cell B5 is trying to divide 192,664 (in cell B2) with a word, which doesn't make sense.

B5 : ✕ ✓ <i>f_x</i> =B2/B3				
	A	B	C	D
1				
2	People	192,664		
3	Houses	Three		
4				
5	Persons/house	#VALUE!		
6				

To fix the error, a value (a number) will need to be entered in cell B3.

➤ #NAME?

This message appears when text is found in a formula that can't be matched to either a legitimate function or range name.

In the example to the left, the formula has been entered as **=SOME(B3:B7)** – there is no such function as SOME, and presumably the author should have typed **=SUM(B3:B7)**.

B9 : ✕ ✓ <i>f_x</i> =SOME(B3:B7)					
	A	B	C	D	E
1					
2		Inventory			
3	Giraffes	34			
4	Tigers	54			
5	Lions	23			
6	Elephants	29			
7	Bats	103			
8					
9	Total	#NAME?			
10					

EXERCISE

1. Which formula below will add the value of B2 to the value of C3

A. =B2+C3

B. =C3+B3

C. =6+4

D. =B+C

Correct Answer

A. =B2+C3

Explanation

The correct answer is =B2+C3 because this formula will add the value of B2 to the value of C3. The plus sign (+) is used to indicate addition in Excel formulas, and the cell references B2 and C3 specify the cells from which the values will be taken. By using this formula, the values of B2 and C3 will be added together to give the desired result.

2. For calculating which of the following options, An Excel spreadsheet is primarily used?

A. Data

B. Finances

C. Numbers

D. All of the above

Correct Answer

D. All of the above

Explanation

An Excel spreadsheet is primarily used for calculating all of the above options, including data, finances, and numbers. Excel provides a wide range of functions and tools that allow users to perform various calculations and analysis on different types of data. It is commonly used in business and finance for tasks such as budgeting, financial analysis, and data management. Additionally, Excel's mathematical functions make it suitable for performing calculations involving numbers. Therefore, all of the given options can be effectively managed and calculated using an Excel spreadsheet.

3. Which is the correct formula?

A. =SUM(A1:A5)

B. Add A1 - A5

C. Subtract the numbers from A1 to A5

D. A1 = A5

Correct Answer

A. =SUM(A1:A5)

Explanation

The correct answer is =SUM(A1:A5). This formula calculates the sum of the values in cells A1 to A5. It adds up all the numbers in this range and gives the total sum as the answer.

4. At the beginning of an Excel spreadsheet formula, the function of the word '=SUM' is

- A. To add all the data together using addition only.
- B. To tell the person viewing that this is a function and it should be added together.
- C. To calculate all the data correctly without any mistakes.
- D. To inform the computer that an arithmetic function will occur.

Correct Answer

A. To add all the data together using addition only.

Explanation

The function of the word '=SUM' at the beginning of an Excel spreadsheet formula is to add all the data together using addition only. This is because the SUM function in Excel is specifically designed to calculate the sum of a range of cells or values. By using the '=SUM' at the beginning of the formula, it indicates to Excel that the following values should be added together.

5. A worksheet is a(n) _____

- A. Piece of graph paper
- B. A single page in a workbook
- C. Excel file
- D. All of the above

Correct Answer

B. A single page in a workbook

Explanation

A worksheet refers to a single page in a workbook. A workbook is a collection of multiple worksheets that are used to organize and store data in Excel. Each worksheet within the workbook functions as an individual page where users can input and manipulate data,

perform calculations, and create charts or graphs. Therefore, the correct answer is "A single page in a workbook."

6. Using a cell address in a formula is known as:

- A. Formularizing
 - B. Prefixing
 - C. Cell referencing
 - D. Cell mathematics
-

Correct Answer

C. Cell referencing

Explanation

Cell referencing is the correct answer because it refers to the practice of using a specific cell address in a formula. By referencing a cell, the formula can retrieve the value or perform calculations based on the data in that particular cell. This allows for dynamic and flexible formulas that can automatically update when the referenced cell's value changes.

7. In which view can you see the header and footer areas of a worksheet

- A. Normal view
 - B. Page layout view
 - C. Page break preview
 - D. Header/Footer
-

Correct Answer

B. Page layout view

Explanation

In Page layout view, you can see the header and footer areas of a worksheet. This view allows you to see how the worksheet will appear when printed, including any headers and footers that have been added. It provides a more accurate representation of the printed document compared to Normal view. In Normal view, the header and footer areas are not visible, and in Page break preview, the focus is on adjusting page breaks rather than viewing headers and footers. The Header/Footer option is not a view but rather a feature that allows you to customize and add content to the header and footer areas.

8. Which of the following is the correct way to write the IF function?

- A. =IF(condition, condition if false, condition if true)
-

- B. =IF(condition, condition if true, condition if false)
- C. =IF(condition:condition if true:condition if false)
- D. None of these

Correct Answer

B. =IF(condition, condition if true, condition if false)

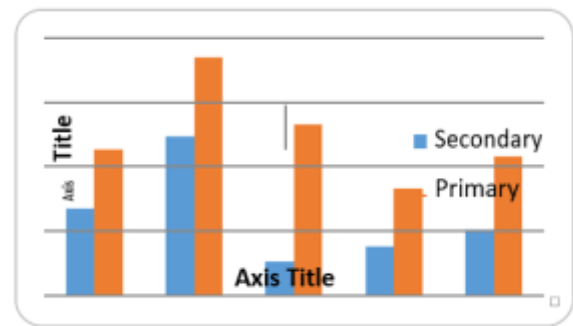
Explanation

The correct way to write the IF function is =IF(condition, condition if true, condition if false). This formula allows you to test a condition and return one value if the condition is true, and another value if the condition is false.

9. Apply a Chart Layout

Since we do not have titles for the chart, x -axis and y-axis therefore by applying layout we can place these titles on the chart.

- a. Click your chart. The Chart Tools become available.
- b. Choose the Design tab.
- c. Select appropriate chart layout (Layout 9) from Chart Layout group.
- d. Excel applies the layout to your chart.
- e. In Chart Title Box, Type Comparison of Schools in Jinja.
- f. In X Axis Title, type District
- a. In Y Axis Title, type No. of Schools
- b. Save the file.

**10. Moving Chart to new sheet**

- a. Click your chart. The Chart Tools become available.
- b. Choose the Design tab.
- c. Click the Move Chart button in the Location group. The Move Chart dialog box appears.
- d. Click the New Sheet radio button.

- e. Type School Comparison - to name the chart sheet. Excel creates a chart sheet named School Comparison and places your chart on it.

11. The following worksheet contains Roll. Nos. & Marks in 5 subject of a student. Calculate his grades using the **HLOOKUP** (Horizontal Lookup) function as per the following:

Marks	Grades
0-40	4
40-50	3
50-60	2
60 & above	1

Student Results

Roll No.	ENG	SST	SCIENCE	MATHS	ICT
110	45	56	67	78	60
GRADE					

Answer

Enter headings & data as shown in columns A to F

To calculate GRADES using HLOOKUP

- ± Enter Marks & Grades in rows 8 & 9 as follows

	A	B	C	D	E
8	MARKS	0	40	50	60
9	GRADE	4	3	2	1

- ± At B3 enter = HLOOKUP(B2, \$B\$8:\$E\$9, 2)
- ± Grade is displayed for the first subject. Drag the formula up to F3 to get grades for all the subjects.

12. The following worksheet contains Names & Sale for 10 salesmen. Calculate their bonus using **VLOOKUP** (Vertical Lookup) as per the following:

Sale	Bonus
0-30000	0
30000-40000	3000
40000-50000	4000
50000-60000	5000
60000-70000	6000
70000-80000	7000

80000 & above	8000
---------------	------

Sales

NAME	SALE	BONUS
Joseph	30000	
Asahel	40000	
Abishag	45000	
Abishai	48000	
Olivia	55000	
Paul	32000	
James	66000	
Maria	23000	
Terry	43000	
Gloria	37000	

Answer

Enter headings & data as shown in columns A to F

To calculate BONUS using VLOOKUP

1. Enter Sale & Bonus in columns H & I as follows

H	I
0	0
30000	3000
40000	4000
50000	5000
60000	6000
70000	7000
80000	8000

2. At C2 enter =VLOOKUP(B2, \$H\$1:\$I\$7, 2)
3. BONUS is displayed for the first salesman. Drag the formula up to C11 to get bonus for all the salesmen.

13. A worksheet contains following data:

	A	B	C	D	E
1	NAME	GENDER	CLASS	CATEGORY	FEES
2	Deep	M	FY	Open	3000
3	Jayesh	M	SY	Reserved	1000

4	Yash	M	TY	Reserved	1000
5	Sara	F	FY	Reserved	500
6	Gita	F	FY	Open	3000
7	Jinal	F	TY	Open	5000
8	Kavita	F	SY	Open	4000
9	Minal	F	SY	Reserved	1000
10	Karan	M	TY	Reserved	1000
11	Abhay	M	TY	Open	5000
12	Bina	F	FY	Open	3000
13	Seema	F	FY	Reserved	500
14	Naresh	M	FY	Reserved	500
15	Rima	F	TY	Open	5000
16	Gajendra	M	SY	Open	4000

Filter the worksheet to show

- Female students from Reserved category
- Male students from TY
- Open category students paying fees > 3000

Answer:

- To display female students from Reserved category
 - Copy paste the header row to some other location in the worksheet say G1
 - Under Gender and Category type F and Reserved respectively
 - Select the entire data A1 : E16
 - From the data tab select Filter Advanced
 - Advanced filter window is displayed
 - Under – Action, select Copy to another location say A20
 - It displays the list range as \$A\$1:\$E\$16
 - Click at criterion range. Select the range you have created along with the header row
 - Click at OK
- To display Male students from TY
 - Copy paste the header row to some other location in the worksheet say G5

- ii. Under Gender and Class type M and TY respectively
- iii. Select the entire data A1 : E16
- iv. From the data tab select Filter Advanced
- v. Advanced filter window is displayed
- vi. Under – Action, select Copy to another location say G20
- vii. It displays the list range as \$A\$1:\$E\$16
- viii. Click at criterion range. Select the range you have created along with the header row
- ix. Click at OK

c) To display Open category students paying fees > 3000

- i. Copy paste the header row to some other location in the worksheet say G10
- ii. Under Category and fees type Open and >3000 respectively
- iii. Select the entire data A1 : E16
- iv. From the data tab select Filter Advanced
- v. Advanced filter window is displayed
- vi. Under – Action, select Copy to another location say A20
- vii. It displays the list range as \$A\$1:\$E\$16
- viii. Click at criterion range. Select the range you have created along with the header row
- ix. Click at OK

DAY #7

UNDERSTANDING QUICK ANALYSIS

The Quick Analysis tools were developed in response to the fact that users weren't using or even aware of the more powerful analytical tools found in Excel.

So Excel decided to combine Live Preview with some of these tools to create the Quick Analysis tools.

➤ The Quick Analysis Button

	A	B	C	D	E	F	G	H	I
1	Alpheius Global Enterprises								
2	Sales								
3		Week 1	Week 2	Week 3	Week 4	Total			
4									
5	Monday	296,114	565,042	429,746	123,445	1,414,347			
6	Tuesday	70,500	78,967	85,889	117,015	352,371			
7	Wednesday	520,830	360,389	244,488	110,585	1,236,292			
8	Thursday	83,296	520,242	82,467	112,728	798,733			
9	Friday	520,140	83,333	87,611	119,158	810,242			
10									
11	Total	1,490,880	1,607,973	930,201	582,931	4,611,985			
12									
13									
14									
15									
16									
17									
18									
19									
20									

FORMATTING

CHARTS

TOTALS

TABLES

SPARKLINES



Data Bars



Color
Scale



Icon Set



Greater
Than



Top 10%



Clear
Format

Conditional Formatting uses rules to highlight interesting data.

The Quick Analysis button appears when a range is selected in a worksheet.

Clicking on the button displays the Quick Analysis gallery which contains quick analysis tools that can be applied to the selected data.

The tools have been organized along tabs at the top – **FORMATTING, CHARTS, TOTALS, TABLES, and SPARKLINES.**

When you click on a tab, options specific to that tab are presented.

➤ Using Quick Analysis Tools with Live Preview

Most of the Quick Analysis tools in the Quick Analysis gallery provide a Live Preview of the changes in the worksheet when you point to an option.

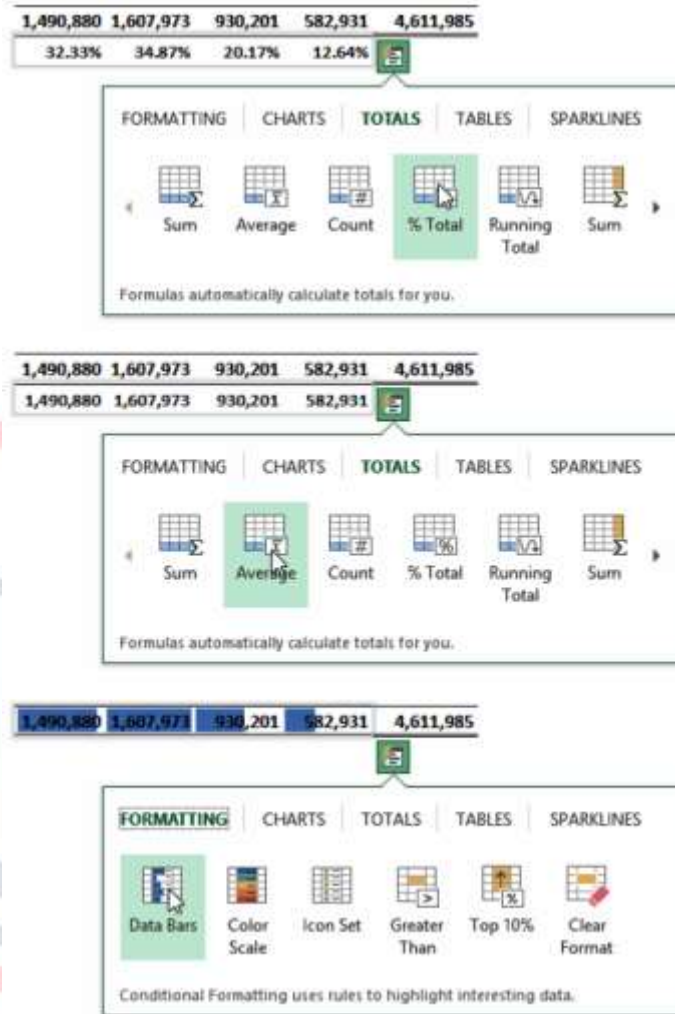
This is very useful if you are not sure of the formatting or type of analysis you require as it provides you with a preview of what the data would look like if you selected that specific option.

At the right we have selected only the totals from the worksheet shown above.

We have pointed to options from the **TOTALS** tab (**% Total** and **Average**) and from the **FORMATTING** tab (**Data Bars**).

Live Preview has either presented another row of analyzed data or has formatted the selection accordingly.

All of these tools are also available on the ribbon but using the **Quick Analysis** tools is much quicker.



QUICK FORMATTING

The first tab in the **Quick Analysis** gallery is **FORMATTING**. This tab provides access to the conditional formatting tools of Excel.

These are the tools that allow you to analyze data by coloring it or presenting it in a slightly different way.

In the **Quick Analysis** gallery, you can apply data bars, color high and low values, values over or below a value, and more.

	A	B	C	D	E	F	G	H	I
1	Alpheius Global Enterprises								
2	Sales								
3		Week 1	Week 2	Week 3	Week 4	Total			
4									
5	Monday	296,114	565,042	429,746	123,445	1,414,347			
6	Tuesday	70,500	78,967	85,889	117,015	352,371			
7	Wednesday	520,830	360,389	244,488	110,585	1,236,292			
8	Thursday	83,296	520,242	82,467	112,728	798,733			
9	Friday	520,140	83,333	87,611	119,158	810,242			
10									
11	Total	1,490,880	1,607,973	930,201	582,931	4,611,985			
12									
13									
14									
15									
16									
17									
18									
19									
20									

FORMATTING

CHARTS

TOTALS

TABLES

SPARKLINES



Data Bars



Color Scale



Icon Set



Greater Than



Top 10%



Clear Format

Conditional Formatting uses rules to highlight interesting data.

QUICK CHARTING

Charts aren't all that difficult to create in Excel, especially with the Recommended Charts feature.

However, deciding what style and type of chart can be daunting.

Fortunately, the Charts tools provide a way of seeing what the different charts will look like without having to first create the chart.



QUICK TOTALS

The **TOTALS tab** in the Quick Analysis gallery has some useful tools and options to help you build your worksheet.

You can use the options to analyze data and perform alternate arithmetic operations (e.g. AVERAGE instead of SUM) or use the options to create the totals and calculations in the first place.

QUICK SPARKLINES

Sparklines are mini charts that are embedded into a worksheet, usually immediately adjacent to the data.

Sparklines are only relatively new in Excel and probably haven't gained the acceptance or understanding that Microsoft would like.

So, you'll now find them in the Quick Analysis tools where you can easily implement them without too much head scratching.

QUICK TABLES

In computer terminology a table is created when data is organized into rows and columns. You'd think then that a worksheet would be a table – but it is not in the Excel definition.

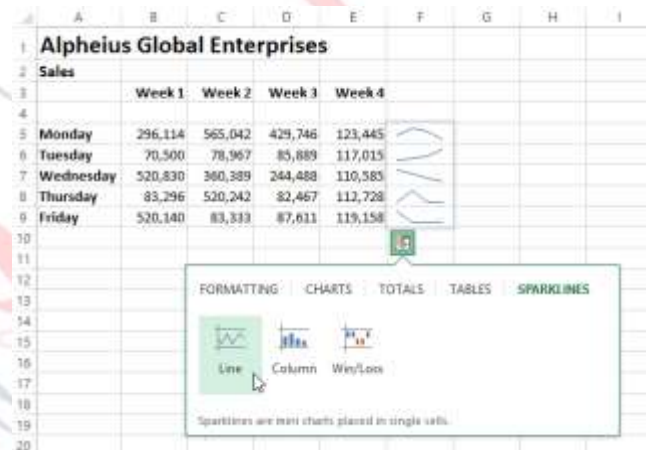
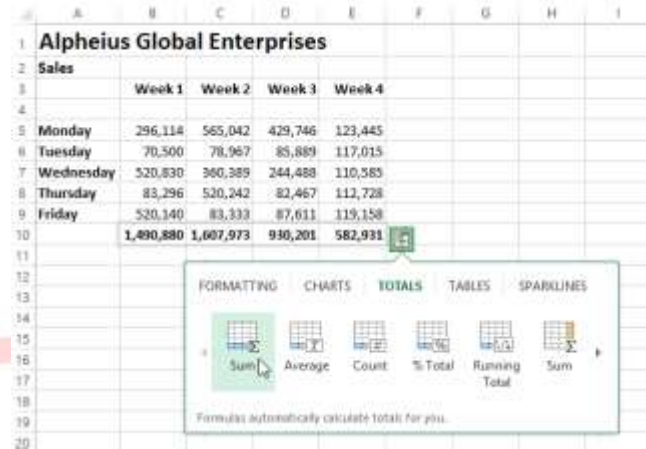
In Excel a table does have columns and rows of continuous data. But it must also have headings which provide filter buttons.

Creating a table is not hard, but it is much easier using Quick Tables.

	A	B	C	D	E	F	G
3	No	First Name	Last Name	Position	Age	Salary	Office
5	NZ0000002	Mary	Campbell	Effective People Leader	56	89,670	Auckland
23	IR0000002	Suzanne	O'Dowd	Effective People Leader	54	98,637	Dublin
41	AU0000002	Harry	Jones	Effective People Leader	37	109,980	Melbourne
59	US0000002	Marty	Zimmstein	Effective People Leader	34	88,773	New York
77	FR0000002	Chantelle	Poiet	Effective People Leader	23	110,967	Paris
94							
95							

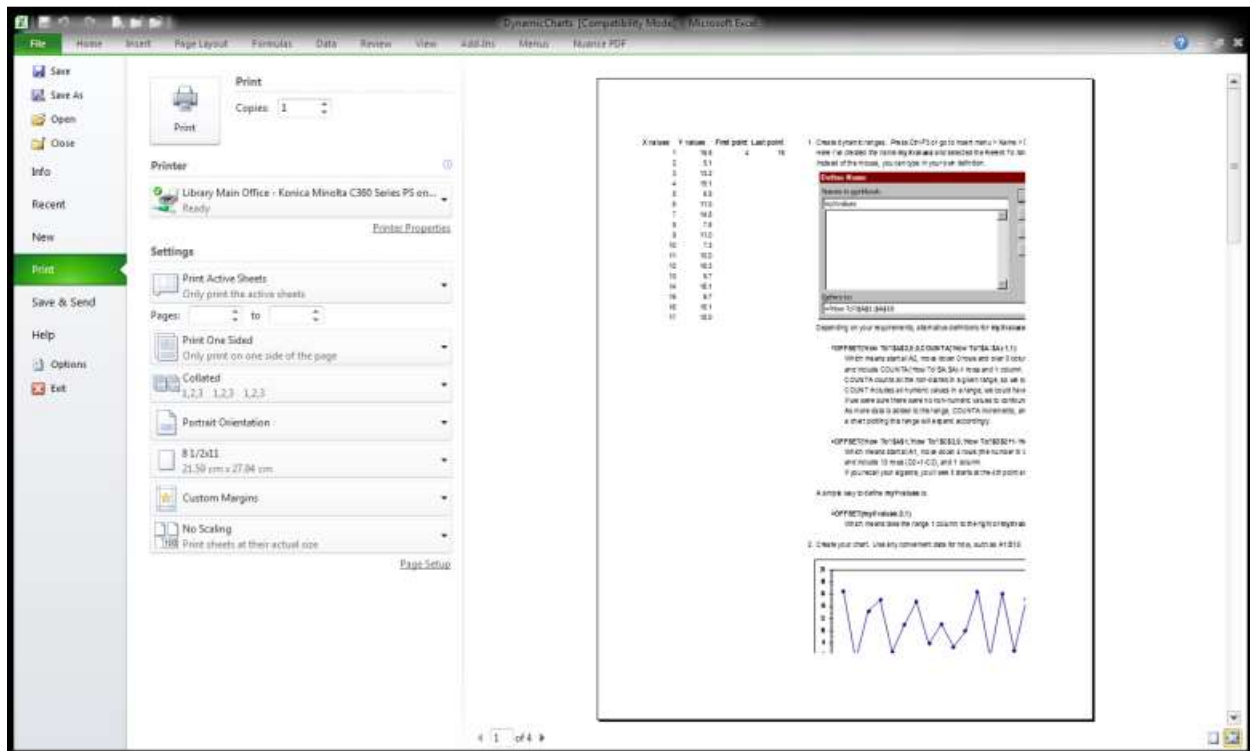
PRINTING A WORKSHEET

Traditionally, printing means producing your document on paper, but in today's Web and online world it might mean printing to the Web or to another file.



Excel gives you a lot of control over what and how much to print, as well as enabling you to select the printer to use.

You can print one or multiple copies of a document, one or multiple pages and even collate copies.



THE CHARTING PROCESS

Charts provide a way of seeing trends in the data in your worksheet. The charting feature in Excel is extremely flexible and powerful and allows you to create a wide range of charts from any of the Insert commands in the Charts group on the:

➤ Inserting Charts

The first step when creating a chart is to select the data from the worksheet that you want to chart. It is important to remember that the selected range (which can be either contiguous or non-contiguous), should include headings (e.g. names of months, countries, departments, etc). These become labels on the chart.

Secondly, the selected range should not (normally) include totals as these are inserted automatically when a chart is created.

The second step is to create a chart using the INSERT tab on the ribbon. You can choose a Recommended Chart where Excel analyses the selected data and suggests several possible chart layouts.

Alternatively, you can create the chart yourself from scratch by choosing one of the Insert commands in the Charts group.

Charts that you create in Excel can be either embedded into a worksheet, or they can exist on their own sheets, known as chart sheets.

➤ Embedded Charts

Charts that appear within a worksheet are known as embedded charts.

A chart is really an object that sits on top of the worksheet – unlike numbers and letters, charts are not actually placed into worksheet cells.

➤ Chart Sheets

If you want to keep your chart separate from the data, you can move the chart to its own sheet.

Chart sheets make it easier and more convenient to work with your chart because you'll see more of it on the screen – since the data is not there!



CHOOSING THE RIGHT CHART

A chart is far more effective at communicating results, outcomes or trends than a table of figures displaying the same information.

Different chart types have been created to communicate different types of information. Some charts show simple relationships between values, while others are designed for quite technical purposes. Here is a summary of the use of different chart types.

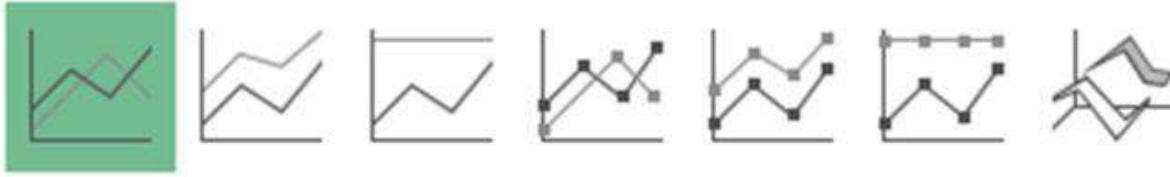
➤ **Column, Bar**

These chart types, either in 2D or 3D, are used to compare values across categories. For example, they could compare the populations of different countries.



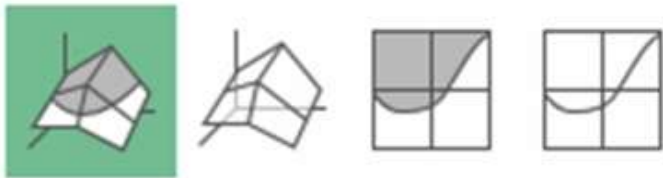
➤ **Line, Area**

Lines in 2D or 3D are useful for showing trends such as sales or employment figures. An area chart is a line chart with the area below the line filled in.



➤ **Surface**

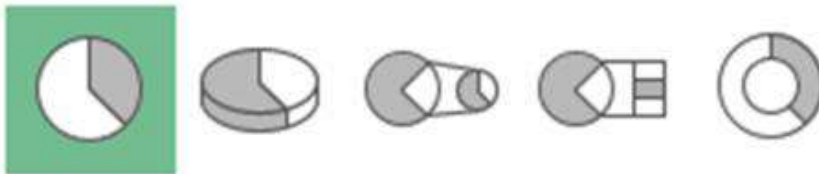
The surface chart plots trends in two dimensions. You could use this to plot departmental sales figures over time. The chart then shows you the trends between departments, as well as the sales trends over time.



➤ **Pie, Doughnut**

If you want to show proportion, such as the sales figures from different departments that make up a total, then the pie and doughnut charts are for you.

The only variation between the doughnut chart and the pie chart is that the doughnut chart can display more than one series of values.



➤ **Stock**

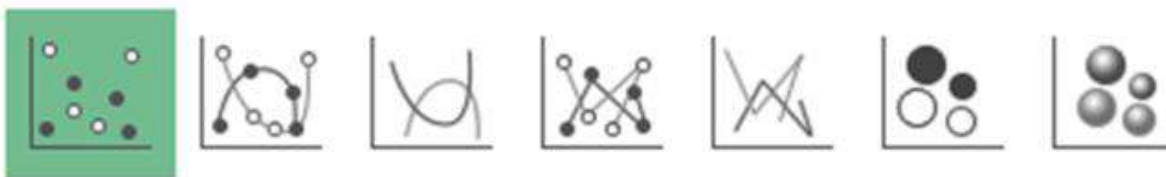
The stock chart type has been designed to show the stock figures for a day, and the trend over time. At its simplest, you can plot the high, low and close figures, and at its most complex, the volume, open, high, low, and close. It can be adapted to show the relationships between any five sets of values.



➤ **XY (Scatter)**

Scatter diagrams are used to display the relationship between two variables. For example, you could research the age and price of a series of cars, and plot the values you find.

You could also investigate the height and weight relationship of a group of people.



➤ Radar

A radar diagram is designed to show the change in values from a central point.

For example, it can be used to show mobile telephone coverage, including multiple networks and multiple measurements.



USING A RECOMMENDED CHART

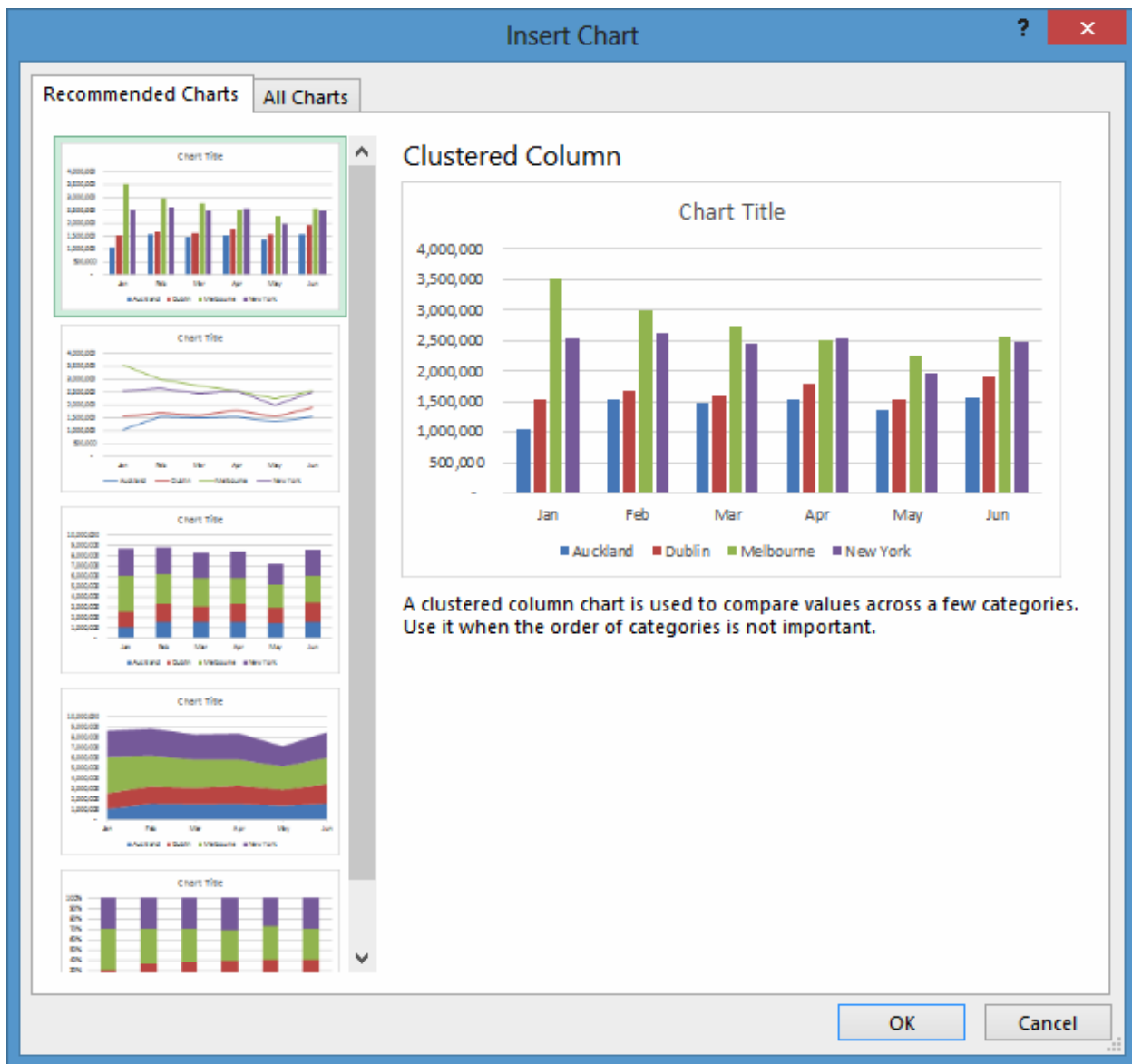
If you are undecided about the best type of chart for the data you have selected to graph, then you may wish to use Excel's Recommended Charts feature.

This feature analyses your selected data and presents you with what it considers to be the best way to chart that data. Several alternatives are presented and you simply choose the one you like most.

	A	B	C	D	E	F	G	H
1	Alpheius Global Enterprises							
2								
3	Revenue	Jan	Feb	Mar	Apr	May	Jun	Total
4	<i>Auckland</i>	1,050,254	1,547,000	1,488,369	1,523,124	1,358,654	1,557,147	8,524,548
5	<i>Dublin</i>	1,524,294	1,685,548	1,599,854	1,789,552	1,542,963	1,896,159	10,038,370
6	<i>Melbourne</i>	3,521,487	2,985,448	2,741,221	2,521,447	2,255,665	2,558,666	16,583,934
7	<i>New York</i>	2,531,225	2,621,889	2,453,999	2,547,441	1,977,558	2,477,332	14,609,444
8								
9	Total Revenue	8,627,260	8,839,885	8,283,443	8,381,564	7,134,840	8,489,304	49,756,296
10								

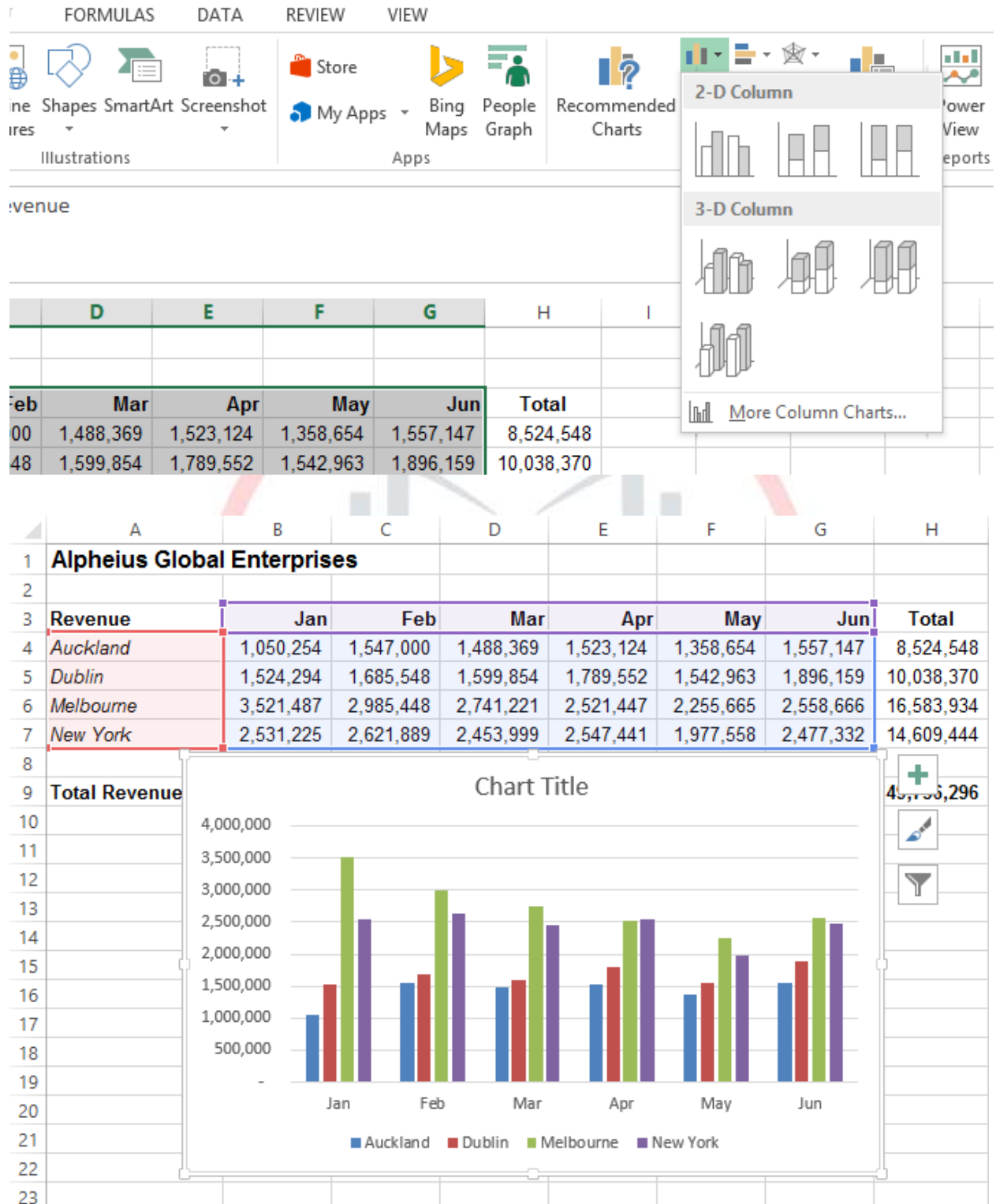
You can also use the Quick Analysis tool that appears at the bottom right corner of a selected range to create a quick chart.

However, this method will not allow you to preview a wide variety of charts.



CREATING A NEW CHART FROM SCRATCH

The easiest way to create a chart is by using the **Recommended Chart** feature. However, you can create a chart yourself from scratch using **INSERT** tab of the ribbon. This may be faster if you have a specific style of chart in mind.

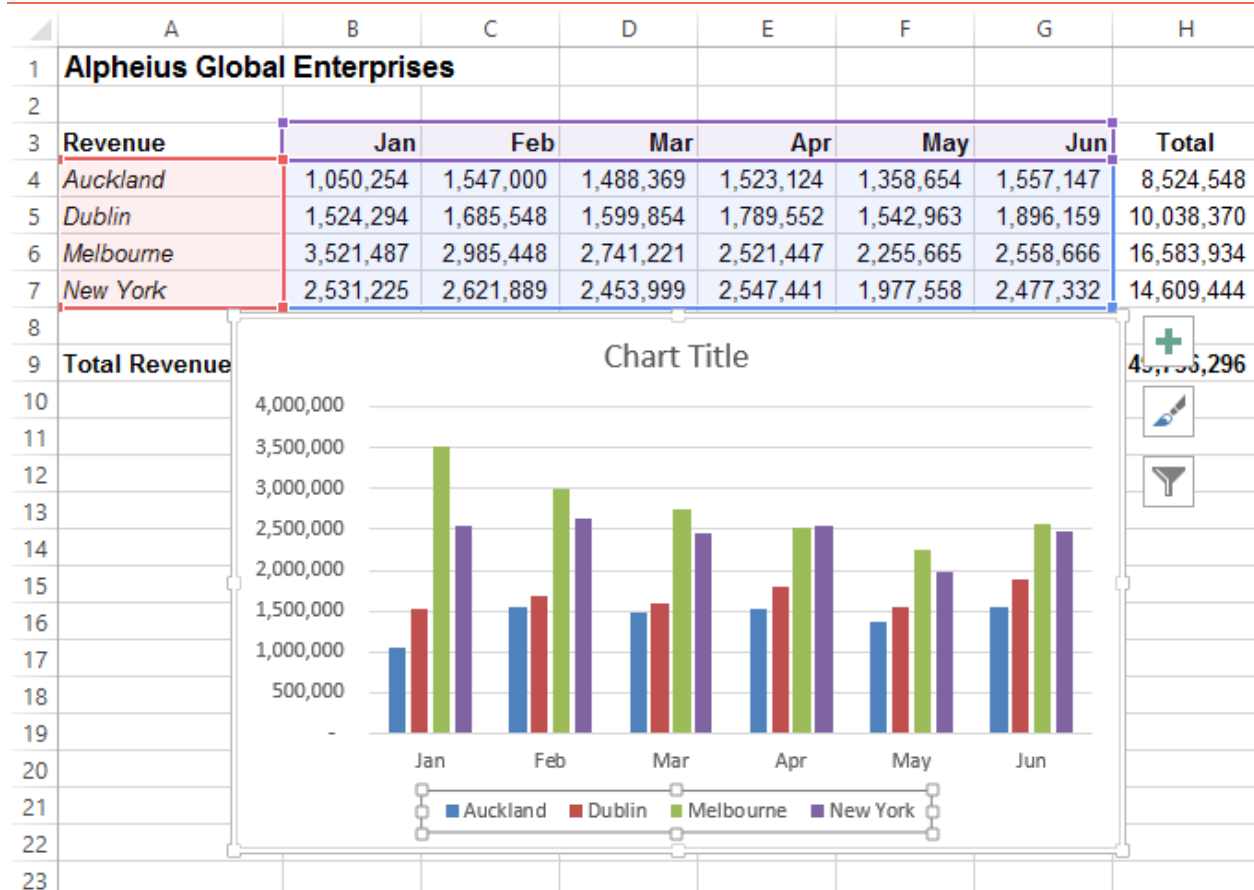


WORKING WITH AN EMBEDDED CHART

By default, new charts are placed in the active worksheet, which is usually the one that contains the data. Charts are placed over the top of the worksheet, embedded as objects.

When you want to work with a chart you must select it – this can be done by clicking on the chart.

The chart itself is made up of many objects and these too can be selected by clicking on them.



RESIZING A CHART

There are two main ways to resize a chart if you are not satisfied with its current size.

A chart that has been selected can be resized by dragging one of the sizing handles around its border.

These handles appear with dots in them. You can also resize a chart using commands in the Size group on the **CHART TOOLS: FORMAT** tab that appears when the chart is selected.

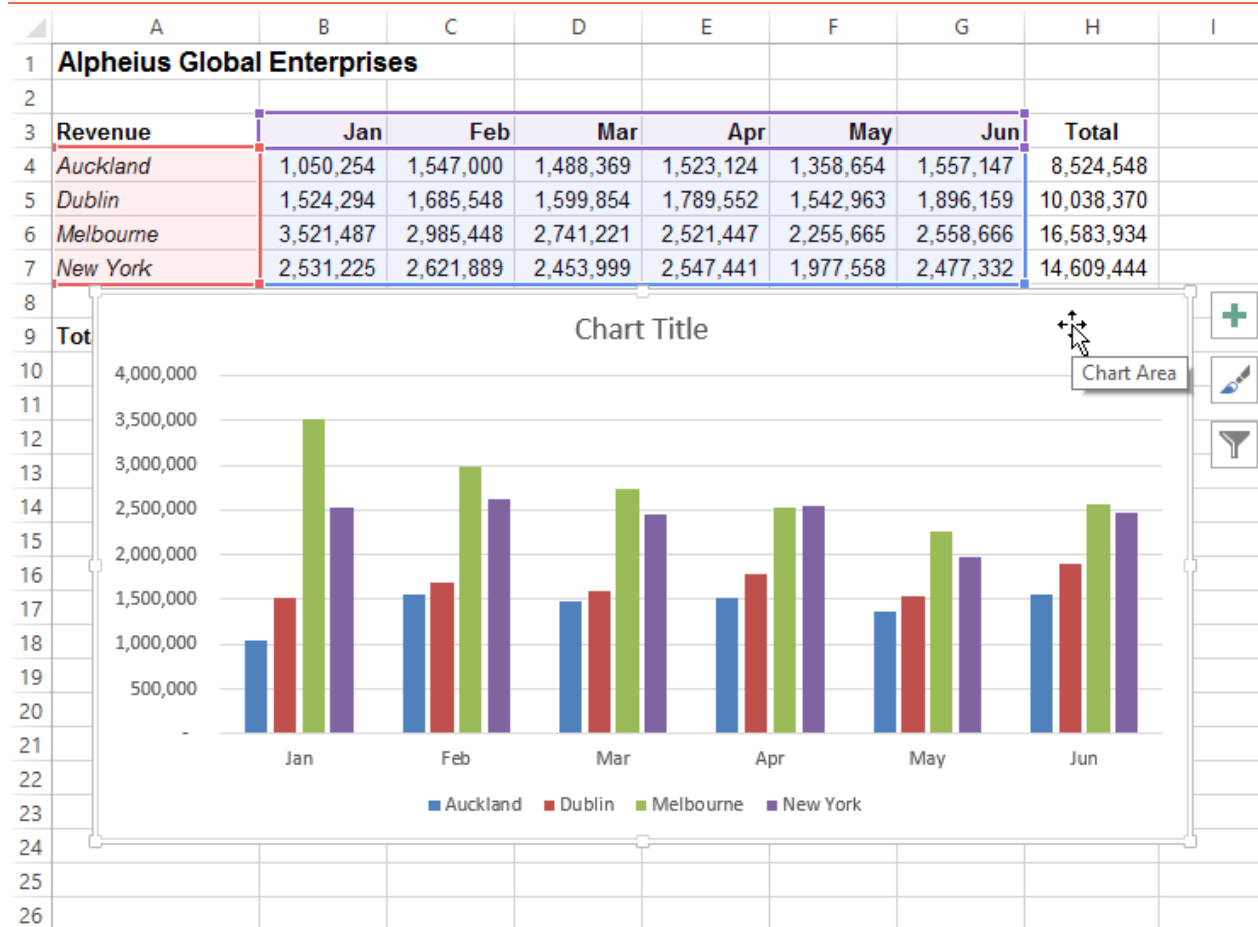


REPOSITIONING A CHART

It's unlikely that a chart embedded in the worksheet by Excel will be exactly where you would like it to be.

You can easily relocate a chart to a more appropriate position by clicking on and dragging the border of the chart to the desired location.

The chart obviously must be selected before it can be dragged to a new position.



PRINTING AN EMBEDDED CHART

When you print a worksheet, Excel will print whatever is in or embedded in that worksheet (including charts).

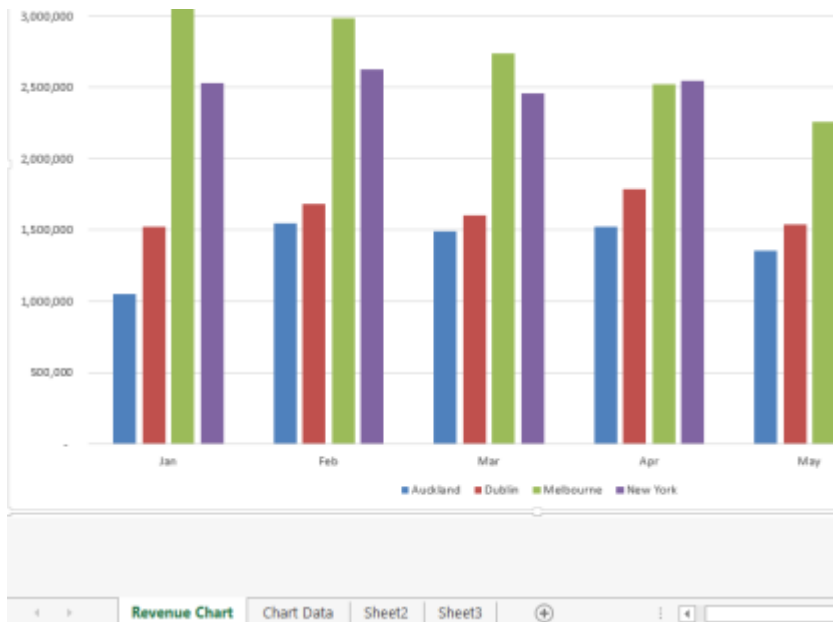
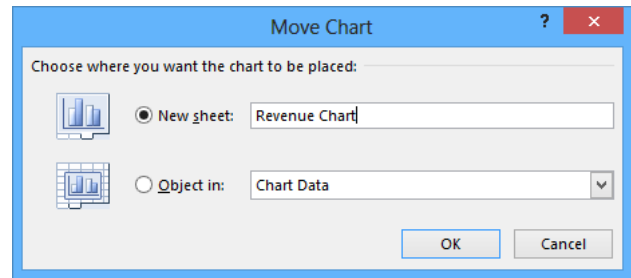
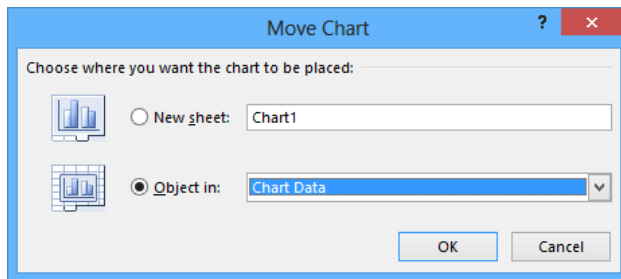
This makes it easy and convenient to print both the chart and its underlying data. All you need to do is to position the chart in the appropriate location then access the print commands in the usual way.



CREATING A CHART SHEET

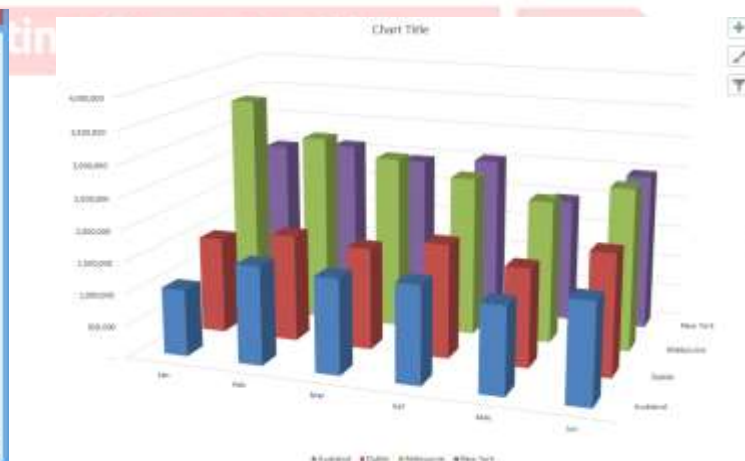
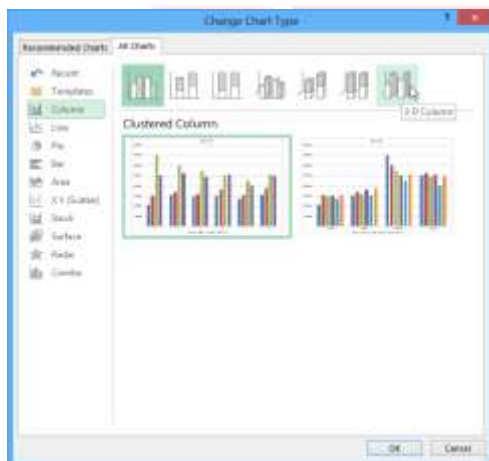
Charts can either be stored in a worksheet or in a separate sheet of their own known as a chart sheet. Chart sheets separate the chart from the underlying data and are useful especially if you are interested in printing the chart on its own page.

Charts can be shifted back and forth between a worksheet and a chart sheet.



CHANGING THE CHART TYPE

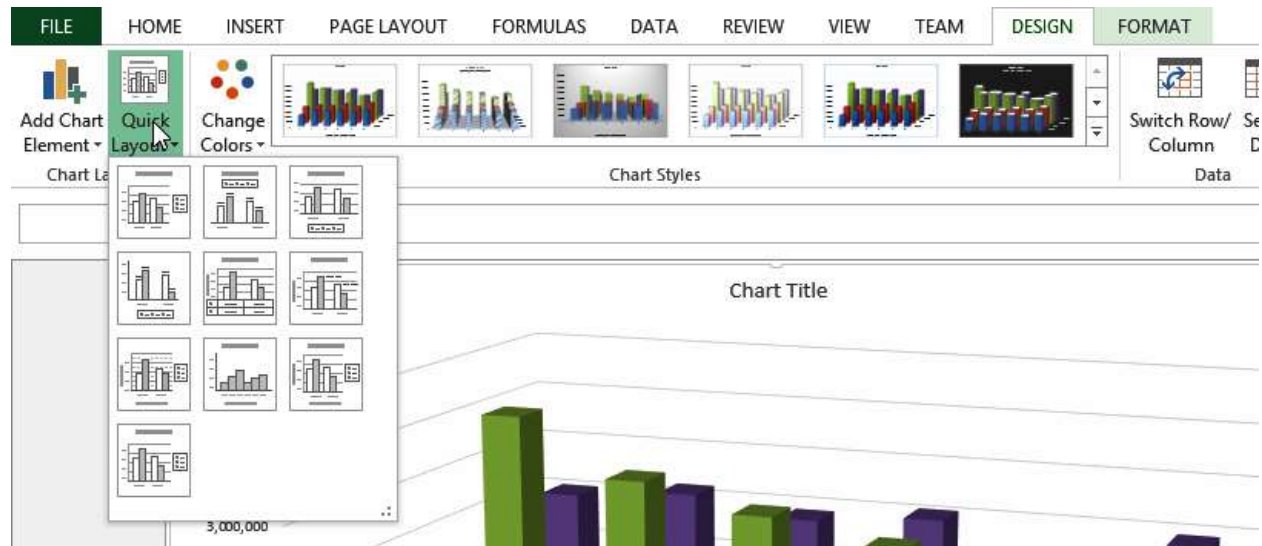
When you create a chart, you may not always achieve the result that you desire. Fortunately, the process for changing a chart type is quite simple. You just need to have an understanding of what each chart type is designed for and to select the format that best suits your purpose. Just be aware that some chart types are designed for specialized applications.



CHANGING THE CHART LAYOUT

Excel has a gallery of chart layouts that can be applied to an existing and selected chart that is either in its own worksheet or embedded into the data worksheet.

Chart layouts are the way elements of the chart are placed within the chart. Different layout options can therefore change the appearance of your chart and its readability.



CHANGING THE CHART STYLE

The style of a chart refers to its colour scheme and overall appearance and can impact the clarity of the content of the chart.

Choosing a predefined chart style can save valuable time and effort. Excel also makes it easy to change chart styles if you decide the style you have chosen is not appropriate.



PRINTING A CHART SHEET

You can print an embedded chart simply by printing the worksheet as if it is a standard worksheet. You can also print a chart sheet in exactly the same way.

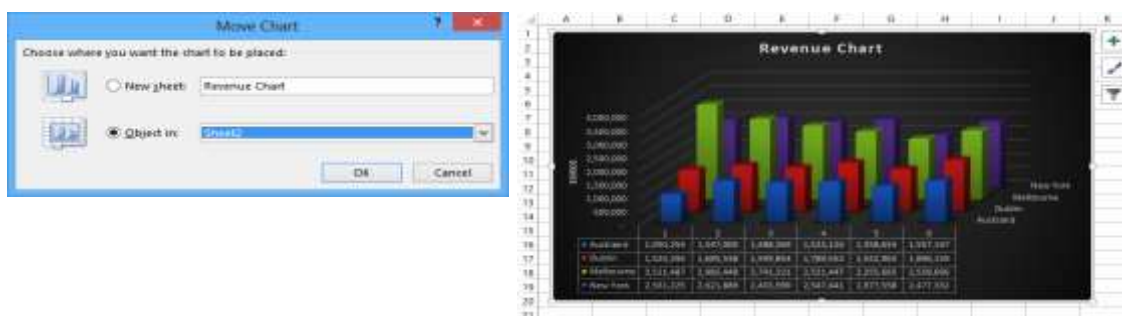
To print a chart sheet, the worksheet data. But the real benefit of inserting simply ensure that the chart sheet is active, then click on the **FILE** tab, click on **Print**, apply the print settings as desired and click on **[Print]**.



EMBEDDING A CHART INTO A WORKSHEET

Charts can either be presented in their own sheets or they can be embedded into a worksheet that contains data.

In fact, you can move a chart back and forth between its own sheet and a worksheet as often as you wish without impacting at all on the chart. Sometimes it is easier to work with a chart in its own sheet, but it may be necessary to print the chart with its data.



DELETING A CHART

If you no longer require a chart you can easily delete it. With embedded charts you must first select the chart in the worksheet and then press the key to delete the chart.

With charts in chart sheets you can delete the sheet by right clicking on the chart sheet tab and choosing the deletion option.

EXERCISE

1. The first step in creating a Pivot Table is:
 - A. Clicking on the Insert Tab and inserting a Pivot Table
 - B. Select data that needs to be analyzed.
 - C. Deciding on which fields to use to analyze the data.
 - D. None of the above
-

Correct Answer

B. Select data that needs to be analyzed.

Explanation

Before you can create a Pivot Table, you must first identify and select the data range you want to analyze. Once the data is selected, you can proceed with inserting a Pivot Table from the Insert tab and determining which fields to use for analysis.

2. Tables created through Excel's table feature allow users to filter ____ by different values.
 - A. Rows
 - B. Columns
 - C. Sections
 - D. Values
-

Correct Answer

B. Columns

Explanation

Tables created through Excel's table feature allow users to filter columns by different values. This means that users can easily select specific columns and apply filters to display only the data that meets certain criteria. This feature is particularly useful when working with large datasets, as it enables users to quickly analyze and manipulate data based on specific column

values. By filtering columns, users can focus on the relevant information and make more informed decisions.

3. ____ tables allow filtering through different values.

- A. Pivot
 - B. Data
 - C. None of them
 - D. Both of them
-

Correct Answer

A. Pivot

Explanation

Pivot tables allow filtering through different values. Pivot tables are a data summarization tool used in spreadsheet programs like Microsoft Excel. They allow users to manipulate and analyze large amounts of data by organizing it into a more manageable format. One of the key features of pivot tables is the ability to filter data based on different values, such as specific dates, categories, or numerical ranges. This allows users to focus on specific subsets of data and gain insights more easily.

4. Creating tables is as easy as highlighting cells that have already been filled in appropriately and then clicking on the insert tab and then clicking on the table button.

- A. No, highlighting cells disables the ability to create tables.
 - B. Yes, it is that simple, but the highlighted cells must have numerical data.
 - C. No, you go to the home tab and then click on the table button.
 - D. No, creating tables in Excel is horribly complicated.
 - E. Yes, it is that simple, but the highlighted cells should not have blank cells.
-

Correct Answer

E. Yes, it is that simple, but the highlighted cells should not have blank cells.

Explanation

The correct answer is "Yes, it is that simple, but the highlighted cells should not have blank cells." This answer explains that creating tables in Excel is indeed easy, but the highlighted cells should not contain any blank cells. This means that all the cells that are being selected to create a table should have data in them, and not be empty.

5. Once data in a table gets filtered, you cannot unfilter the table nor get the table back to the original settings. Is this statement correct?

- A. No, you may change filters up to three times.
 - B. Yes
 - C. No, you may change filters as many times as you please without limits
 - D. No, tables cannot be filtered.
-

Correct Answer

C. No, you may change filters as many times as you please without limits

Explanation

The statement provided in the question is incorrect. In most table filtering options, you can change filters as many times as you want without any limits. This means that you can apply different filters, remove them, and apply new ones multiple times to manipulate the data in the table according to your requirements. So, the correct answer is that you can change filters as many times as you please without any limitations.

6. Which pivot table layout allows dragging fields to the grid in a spreadsheet?

- A. Default Pivot Table Layout
 - B. Classic Pivot Table Layout
 - C. Master Pivot Table Layout
 - D. Operation Geronimo Layout
-

Correct Answer

B. Classic Pivot Table Layout

Explanation

The Classic Pivot Table Layout allows dragging fields to the grid in a spreadsheet. This layout is a traditional and widely used format for creating pivot tables. It provides a user-friendly interface that allows users to easily manipulate and organize data by dragging and dropping fields into the desired areas of the grid. The Classic Pivot Table Layout is a popular choice for creating dynamic and interactive reports and analysis in spreadsheet applications.

7. For a table to work properly, the top row should have column headings. Yes or no?

- A. Yes
 - B. No, only rows may have headings.
-

- C. No
- D. Can't say

Correct Answer

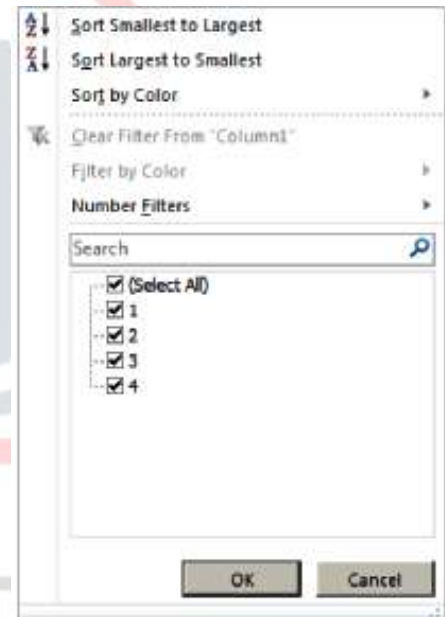
A. Yes

Explanation

In order for a table to work properly, the top row should have column headings. Column headings help to identify and label the data in each column, making it easier for users to understand and interpret the information in the table. Without column headings, it would be difficult to differentiate between different columns and make sense of the data presented in the table. Therefore, it is necessary for the top row of a table to contain column headings.

8. Check all that are true.

- A. The boxes that are checked indicate that the table should filter out that data.
- B. The boxes that are checked indicate that the table will filter in that data.
- C. The boxes that are unchecked indicate that the table will filter out that data.
- D. The boxes that are unchecked indicate that the table will not filter in that data unless values are present.



Correct Answer(s)

- B. The boxes that are checked indicate that the table will filter in that data.
- C. The boxes that are unchecked indicate that the table will filter out that data.

Explanation

The explanation for the given correct answer is that when a box is checked, it means that the table will include or filter in the data that corresponds to that box. On the other hand, when a box is unchecked, it means that the table will exclude or filter out the data that corresponds to that box. This implies that the checked boxes indicate the data that will be included, while the unchecked boxes indicate the data that will be excluded from the table's filtering process.

- 9. Pivot tables that have been created can easily be turned into graphs or charts by simply highlighting the whole pivot table and choosing a graph or chart that you want to create. Is this true?

- A. True
 - B. False, pivot tables do not contain enough information for graphs or charts
 - C. False, only regular tables may be used to create graphs or charts
 - D. False
-

Correct Answer

A. True

Explanation

Pivot tables can indeed be easily turned into graphs or charts by highlighting the entire pivot table and selecting the desired graph or chart format. This is a true statement because pivot tables are designed to summarize and analyze data in a way that makes it easy to visualize patterns and trends. By converting a pivot table into a graph or chart, users can further enhance their understanding of the data and communicate it more effectively.

10. Which of these is NOT included in the PivotTable Field List pane?

- A. Column Labels
 - B. Values
 - C. Report Filter
 - D. Formulas
-

Correct Answer

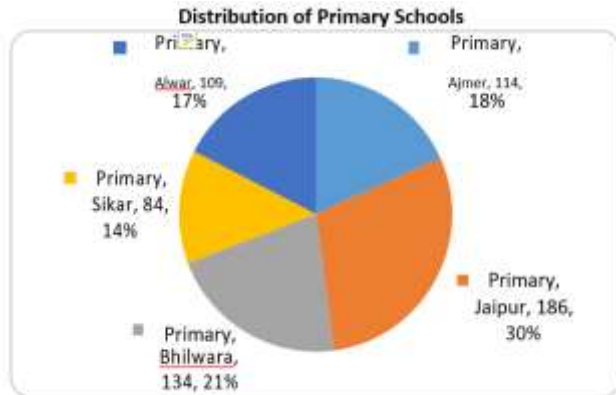
D. Formulas

Explanation

The PivotTable Field List pane in Microsoft Excel is a tool used to organize and manipulate data in a PivotTable. It allows users to drag and drop fields into different areas of the PivotTable to create a customized view. The options included in the PivotTable Field List pane are Column Labels, Values, and Report Filter. However, Formulas are not included in this pane. Formulas are used to perform calculations within the PivotTable, but they are not listed as a separate category in the Field List pane.

11. Prepare a Pie Chart showing % of primary schools in each of the district of the total primary school.

- The Title of the Graph should be Distribution of Primary Schools.
- The Chart should be placed on a separate sheet called Primary School
- The Category Data and % should be displayed outside the chart. (Layout – Data Labels - More Label Options)



12. The following worksheet contains Customer No., Number of units consumed for 10 customers. Calculate their bill amount using **LOOKUP** as per the following:

Number of units	Rate	Number of units	Rate
<200		UGX3	
>=200, < 500		UGX6	
>= 500		UGX10	

Customer No.	No. of Units	Rate	Bill Amount
1101	340		
1102	180		
1103	400		
1104	600		
1105	350		
1106	470		
1107	890		
1108	200		
1109	500		
1110	360		

13. A worksheet contains Roll Number, Marks in 2 subjects for 10 students in a class. Calculate Result and Grade using the following:

- A student is declared as PASS if he gets 40 or more in both the subjects, Otherwise FAIL.
- All FAILED students will be given Grade IV
- For PASSED students Grade will be obtained as follows:

AVERAGE GRADE	AVERAGE GRADE
>=60	I
<60 but >=50	II
<50 but >=40	III

Students Results

Roll No.	Math	English	Average	Result	Grade
1101	34	80			
1102	18	41			
1103	40	65			
1104	60	77			
1105	35	78			
1106	47	87			
1107	89	32			
1108	20	78			
1109	50	65			
1110	36	55			

14. The following worksheet contains Name & Sales of 10 salesmen. Calculate commission as per the following:

Sales	Commission
First 30,000	5%
Next 40,000	10%
Excess	15%

Salesmen Commissions

	A	B	C
1	NAME	SALE	COMMISSION
2	Joseph	30000	
	Asahel	40000	
	Abishag	45000	
	Abishai	48000	
	Olivia	55000	
	Paul	32000	
	James	66000	
	Maria	23000	
	Terry	43000	
	Gloria	37000	

Tip:

= IF(B2 <30000, B2*5%,IF(B2<=70000,1500+(B2-30000)*10%, 1500 +4000+ (B2-70000)*15%))

DAY #8

EXERCISE

14. To add a new row, click a cell in the row immediately above where you want the new row.

A. True

B. False

Correct Answer

B. False

Explanation

To add a new row in a table, you need to click on a cell in the row immediately below where you want the new row to be inserted. Therefore, the statement "To add a new row, click a cell in the row immediately above where you want the new row" is false.

15. To copy an entire worksheet and all its data, you should click the worksheet tab of the sheet that you want to copy, hold down SHIFT, and then drag the selected sheet along the row of sheet tabs.

A. True

B. False

Correct Answer

B. False

Explanation

To copy an entire worksheet and all its data, you should click the worksheet tab of the sheet that you want to copy, hold down CTRL, and then drag the selected sheet along the row of sheet tabs.

16. If cells: A1=90 A2=85 A3=80 A4=75 A5=75 What will be your formula if you are going to get the average?

Correct Answer

=AVERAGE(A1:A5)

=(A1+A2+A3+A4+A5)/5

=average(a1:a5)

=(a1+a2+a3+a4+a5)/5

Explanation

The formula to calculate the average of the given cells is =AVERAGE(A1:A5) or =(A1+A2+A3+A4+A5)/5 or =average(a1:a5) or =(a1+a2+a3+a4+a5)/5.

17. If cells: A1=90 A2=85 A3=80 A4=75 A5=75 What will be your formula if you are going to get the total?

Correct Answer

=SUM(A1:A5)

=A1+A2+A3+A4+A5

=sum(a1:a5)

=a1+a2+a3+a4+a5

Explanation

The correct formula to get the total of the given cells is =SUM(A1:A5) or =A1+A2+A3+A4+A5 or =sum(a1:a5) or =a1+a2+a3+a4+a5. These formulas will add up the values in cells A1 to A5 and give the total sum.

18. Get the remarks in cell E10 that valued 75 which noted, if the remarks will be higher than 74, the remarks will be "PASSED" or else "FAILED".

Correct Answer

=IF(E10>74,"PASSED","FAILED")

Explanation

The given answer is a formula in Excel that uses the IF function. It checks if the value in cell E10 is greater than 74. If it is, the formula returns "PASSED". If it is not, the formula returns "FAILED". This formula is used to determine whether a student has passed or failed based on their remarks.

19. Get the remarks in cell A1 that valued 75 which noted, if the remarks will be higher than 74, the remarks will be "PASSED" or else "FAILED".

Correct Answer

=IF(A1>74,"PASSED","FAILED")

Explanation

The correct answer is the formula =IF(A1>74,"PASSED","FAILED"). This formula uses the IF function to check if the value in cell A1 is greater than 74. If it is, the formula returns "PASSED". If it is not, the formula returns "FAILED". This formula is used to determine whether the remarks should be "PASSED" or "FAILED" based on the value in cell A1.

20. A worksheet contains name and marks in 3 subjects. Calculate Total Marks

	A	B	C	D	E
1	NAME	SUB 1	SUB 2	SUB 3	TOTAL MARKS
2	Deep	30	34	44	
3	Jayesh	40	35	45	
4	Yash	45	36	47	

5	Sara	48	32	50	
6	Gita	35	32	43	
7	Jinal	32	31	37	
8	Kavita	36	28	38	
9	Minal	23	25	40	
10	Nareesh	43	27	50	
11	Rima	37	44	46	

- Construct 3D Pie Chart for Total marks
- Construct 2D Line Chart for Subject 1 and Subject 3
- Construct 2D Column Chart for Sub1, Sub2, Sub3
- Construct Stacked Column Chart for Sub1, Sub2, Sub3

Answers:**3D Pie Chart for Total marks**

- Select entire column of Total Marks ie E1:E11
- Insert Tab—Chart group – Pie option—3D Pie Chart
- 3D Pie diagram appears
- Click at Chart tools – layout– Chart Title – Above the chart It displays a box above the chart. Click inside the box and type the title as **3D PIE CHART FOR TOTAL MARKS**
- Click on the Legend area, right Click and Click at select Data
- Click on Edit at Horizontal Axis Labels
- Select range as A2:A11 (Click and drag) and OK
- Names of students will be displayed in the legend area

2D Line Chart for Subject 1 and Subject 3

- Select B1:B11 Keep pressing Cntrl Key & Select D1:D11
- Insert Tab—Chart group – Line option— sub option 2D
- 2D Line Chart appears
- Click at Chart tools – layout– Chart Title – Above the chart It displays a box above the chart. Click inside the box and type the title as LINE CHART FOR SUBJECT 1 AND 3
- Under layout menu of Chart tools, select Axis Title - sub option Primary horizontal axis – and position Below the Axis

Axis Title box appears below the horizontal axis

Click inside the box and type the title as Student Names

- vi. Under layout menu of Chart tools, select Axis Title - sub option Primary vertical axis – and position Rotated

Axis Title box appears at the vertical axis in rotated position Click inside the box and type the title as Marks

- vii. Click on labels at Horizontal Axis, Right Click and select Data Labels option

Click on Edit at Horizontal Axis Labels

It asks for the range for Horizontal Axis Labels Select the range as A2:A11

Names of students will be displayed in the Horizontal Axis area

OK

2D Column Chart

- i. Select A1: D11
- ii. Insert Tab—Chart group – Column option— sub option 2D
- iii. 2D Column Chart appears
- iv. Under layout menu of Chart tools, select Chart Title , sub option Above the chart
- v. It displays a box above the chart.

Click inside the box and type the title as

COLUMN CHART FOR SUBJECT 1 ,2 AND 3

- vi. Also specify the axis titles as explained above
- vii. Also specify the Horizontal axis labels as explained above

2D Stacked Column Chart

- i. Select A1: D11
- ii. Insert Tab—Chart group – Column option— sub option 2D – Stacked Column
- iii. 2D Stacked Column Chart appears
- iv. Under layout menu of Chart tools, select Chart Title , sub option Above the chart Click inside the box and type the title as

STACKED COLUMN CHART FOR SUBJECT 1 ,2 AND 3

- v. Also specify the axis titles as explained above
- vi. Also specify the Horizontal axis labels as explained above

21. For the following worksheet containing amount spent for various items during the year, prepare scenarios where

- a. Machinery increases to 80,000 , carriage increases to 9000 & Postage increases to 8000
- b. Carriage increases to 10,000 Office equipment increases to 7000 and postage increases to 9000

	A	B
1	Items	Costs
2	Machinery	60000
3	Carriage	8000
4	Transport	30000
5	Office equipment	6000
6	Postage	7000
7	Miscellaneous	3000
8	Generator	5000
9	Total	119000

Answer:

- i. Select Data tab – What if Analysis – Scenario Manager
- ii. Scenario Manager dialogue box appears
- iii. Click on Add in Scenario Manager

Add Scenario dialog box appears

At scenario name -type name as Current Expenses

In changing sales -type B2:B8 – OK

- iv. Click on Add in Scenario Manager

Add Scenario dialog box appears

At scenario name -type Increase in Machinery and carriage

Click on OK

- v. At Scenario Values dialog box , type

For Machinery Text box at \$B\$2 type 80000

For Carriage Text box at \$B\$3 type 9000

For Postage Text box at \$B\$6 type 8000

Click on OK

- vi. Click on Add in Scenario Manager

Add Scenario dialog box appears

At scenario name -type Increase in carriage and office equipment

Click on OK

- vii. At Scenario Values dialog box, type

For Carriage Text box at \$B\$3 type 10000

For office equipment Text box at \$B\$5 type 7000

For Postage Text box at \$B\$6 type 9000

Click on OK

- viii. At Scenario Manager dialog box , click on Summary

- ix. At Scenario Summary dialog box , click on Scenario Summary

Type B9 in Result Cell

- x. Scenarios are displayed on a new scenario worksheet

22. For the following worksheet obtain the solution for the cost price so that the profit will be 20000

	A	B	C	D
1	CP	ADVT	SP	PROFIT
2	9000	900	22000	
3				

Answer:

- i. At D2 enter formula =C2-A2-B2

- ii. Click at D2

iii. Click on Data--- Data Tools group --- What if analysis Subgroup ---

Goal Seek option

iv. Goal Seek dialogue box appears

v. Enter the following data at the Goal seek Dialogue box

Set Cell \$D\$2

To value 20000

vi. It displays Goal Seek status as follows Goal seeking with cell D2

Found a solution Target value

20000 Current value 20000

OK

vii. It displays the output

23. Maximize the profit for the following

Current selling price of the bridal costume is UGX22000, the cost price is UGX10,000 The advertising expenses are UGX500.

The constraints are:

The cost budget should be between UGX9000 and UGX 12000 & the advertising expenditure ranges between UGX 400 and UGX 1000

Answer:

Type the data as follows

	A	B	C	D
1	CP	ADVT	SP	PROFIT
2	10000	500	22000	
3				

i. Click on D2- Data---Analysis group --- Solver

ii. Solver Parameters dialogue box appears. Enter the value as below

Set objectives \$D\$2

To Max

By changing variable cells \$A\$2:\$B\$2

Subject to constraints - Type the constraints one by one by clicking on Add

When done Click on Solve.

- iii. It displays solver results dialogue box

Click at Save scenario

It asks you to enter name for

scenario Enter S1

OK

- iv. It displays the value. It also generates the reports

- v. To view Answer Report

At Solver results dialogue box

Select Answer at Reports Tab

OK

24. The following worksheet contains Name & Taxable Income for 50 employees. Calculate Income Tax Surcharge and Total Tax for the following worksheet

	A	B	C	D	E
1	NAME	TAXABLE INCOME	INCOME TAX	SURCHARGE	TOTAL TAX
2					
:					
:					
50					

Income Tax is calculated as follows :

Taxable Income	Income tax
First 1,50,000	Nil
Next 1,00,000	10%
Next 75,000	20%
Excess	30%

Surcharge is 3% on Income Tax if Taxable income is above 5,00,000

Answer:

Enter headings and data as shown above

To calculate INCOME TAX

In cell C2 enter the formula

`IF(B2 <150000, 0,IF(B2<=250000,(B2-150000)*10%,IF(B2<=325000, 10000 + (B2-250000)*20%,25000+(B2-325000)*30%)))`

and press enter key

It shows Income Tax for the first employee

Drag the formula up to C51

To calculate SURCHARGE

In cell D2 enter the formula = `IF(B2 <500000, 0,C2*3%)` and press enter key It shows Surcharge for the first employee Drag the formula up to D51

To calculate TOTAL TAX

In cell E2 enter the formula = `C2+D2` and press enter key

It shows Total Tax for the first employee

Drag the formula up to E51

